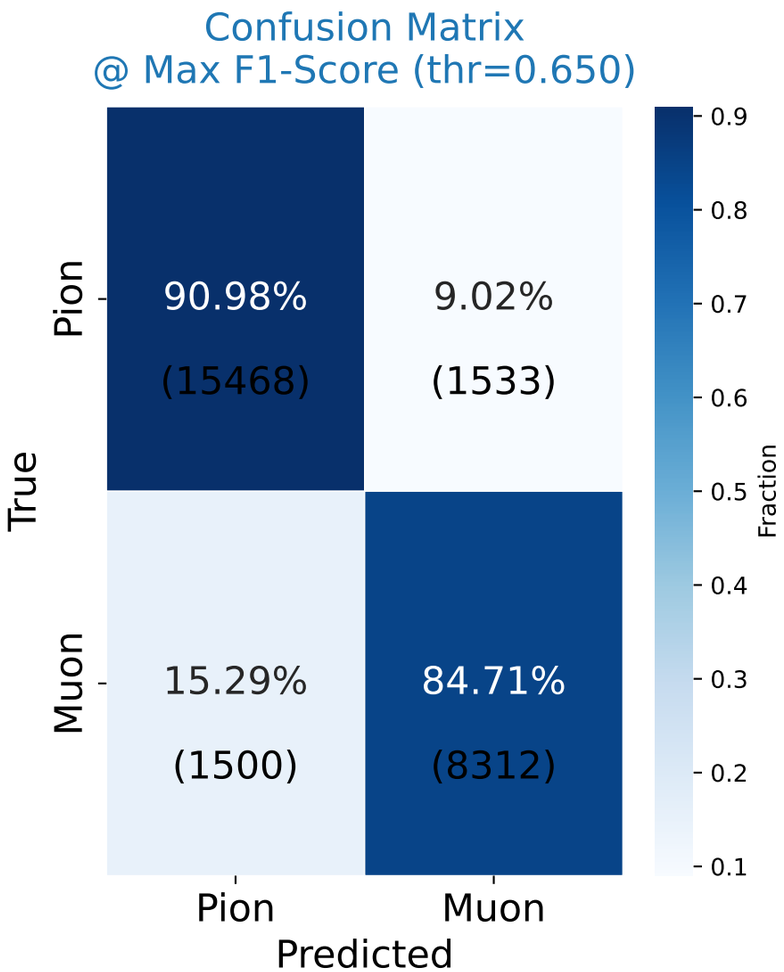
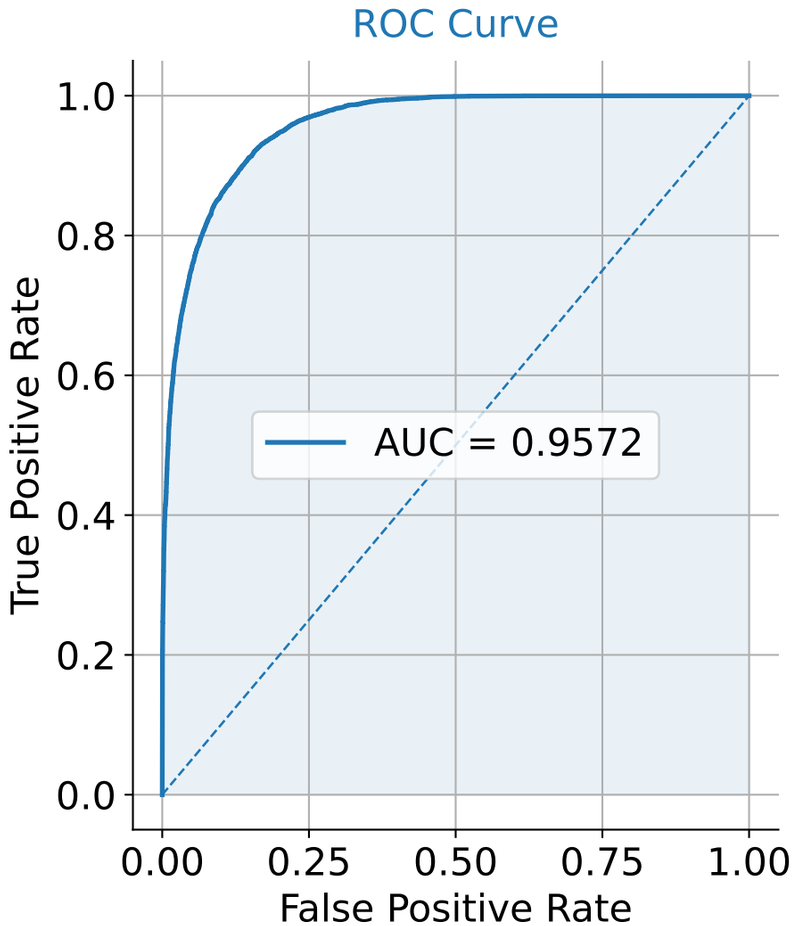
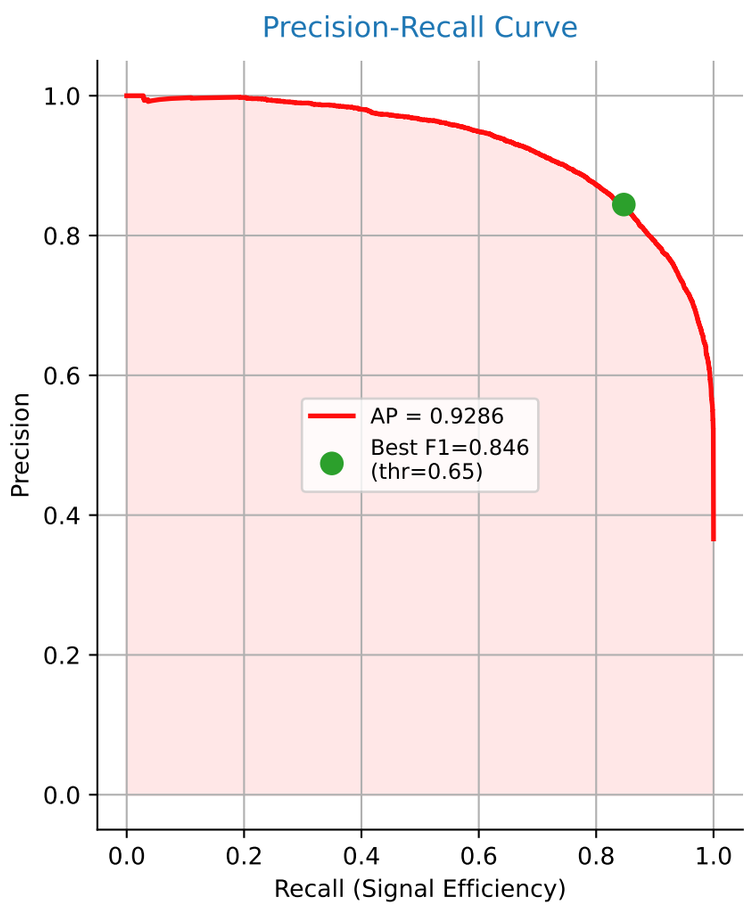
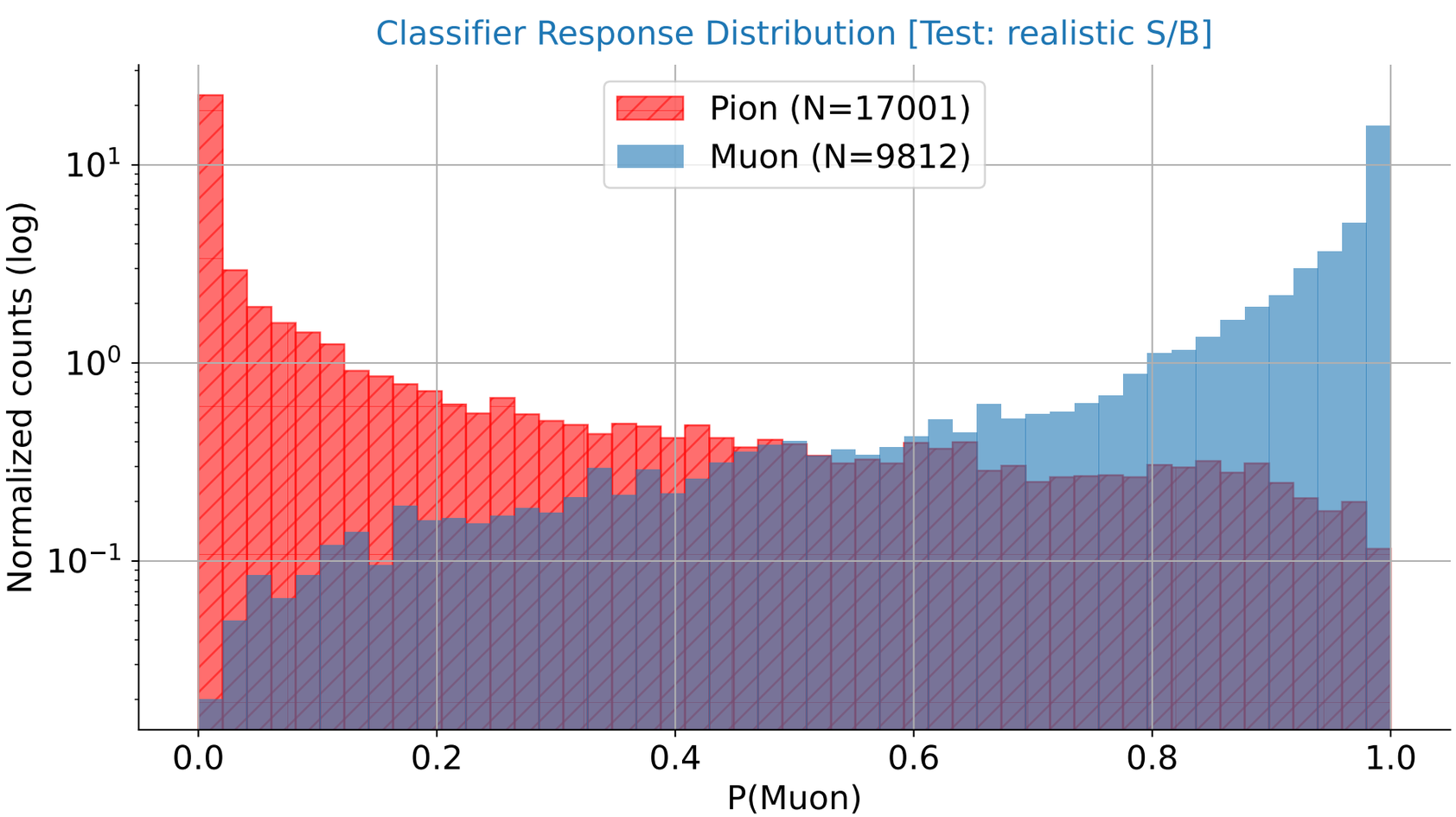
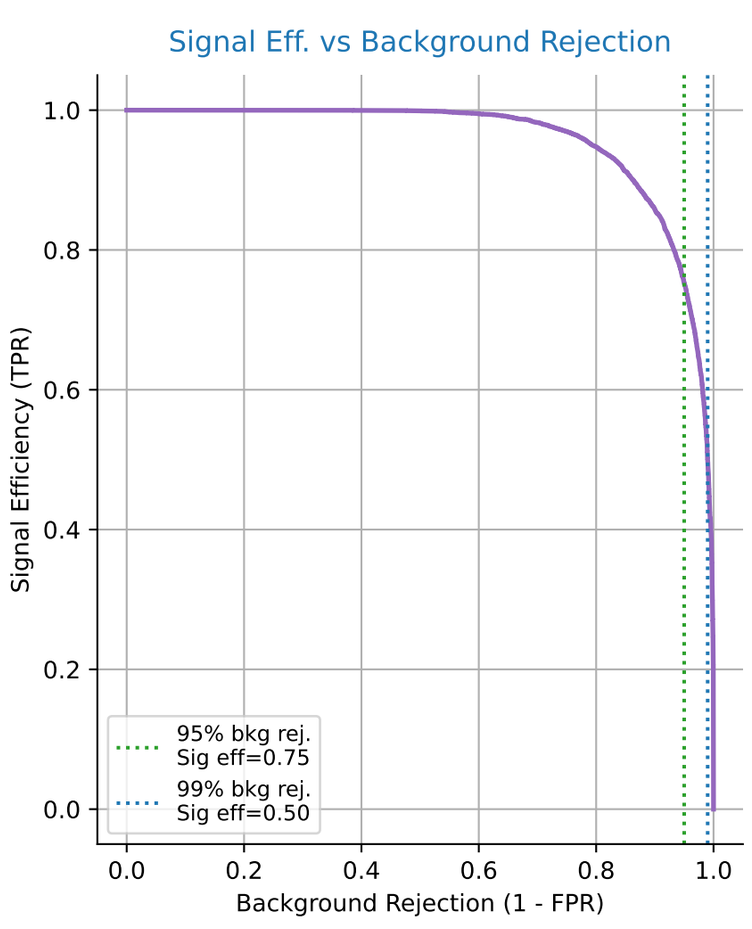
Muon vs Pion ID — XGBoost (Calorimeter + ToF features)
[Train S/B=1:2 | Test S/B=1:2 (realistic, all files)]



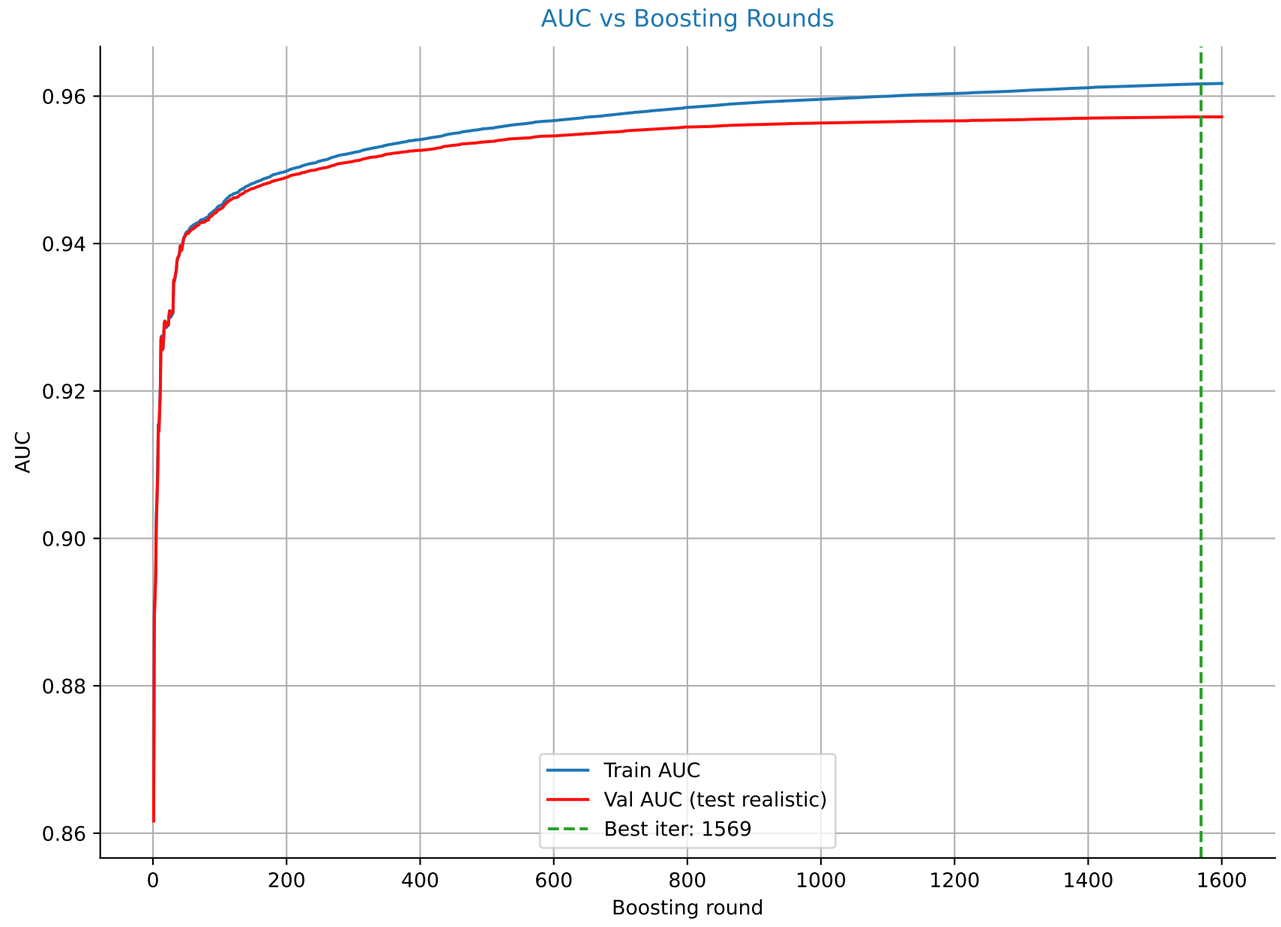


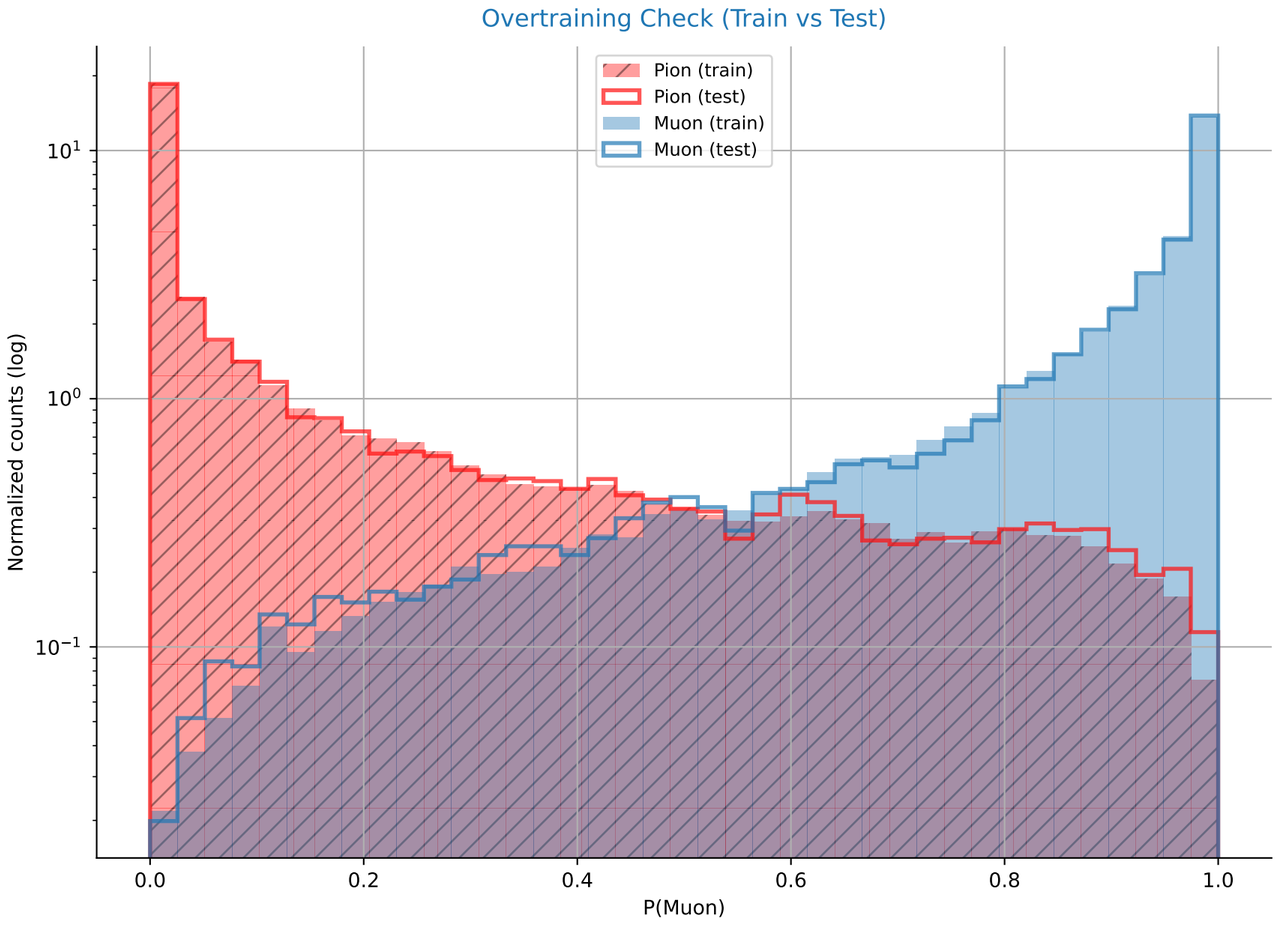






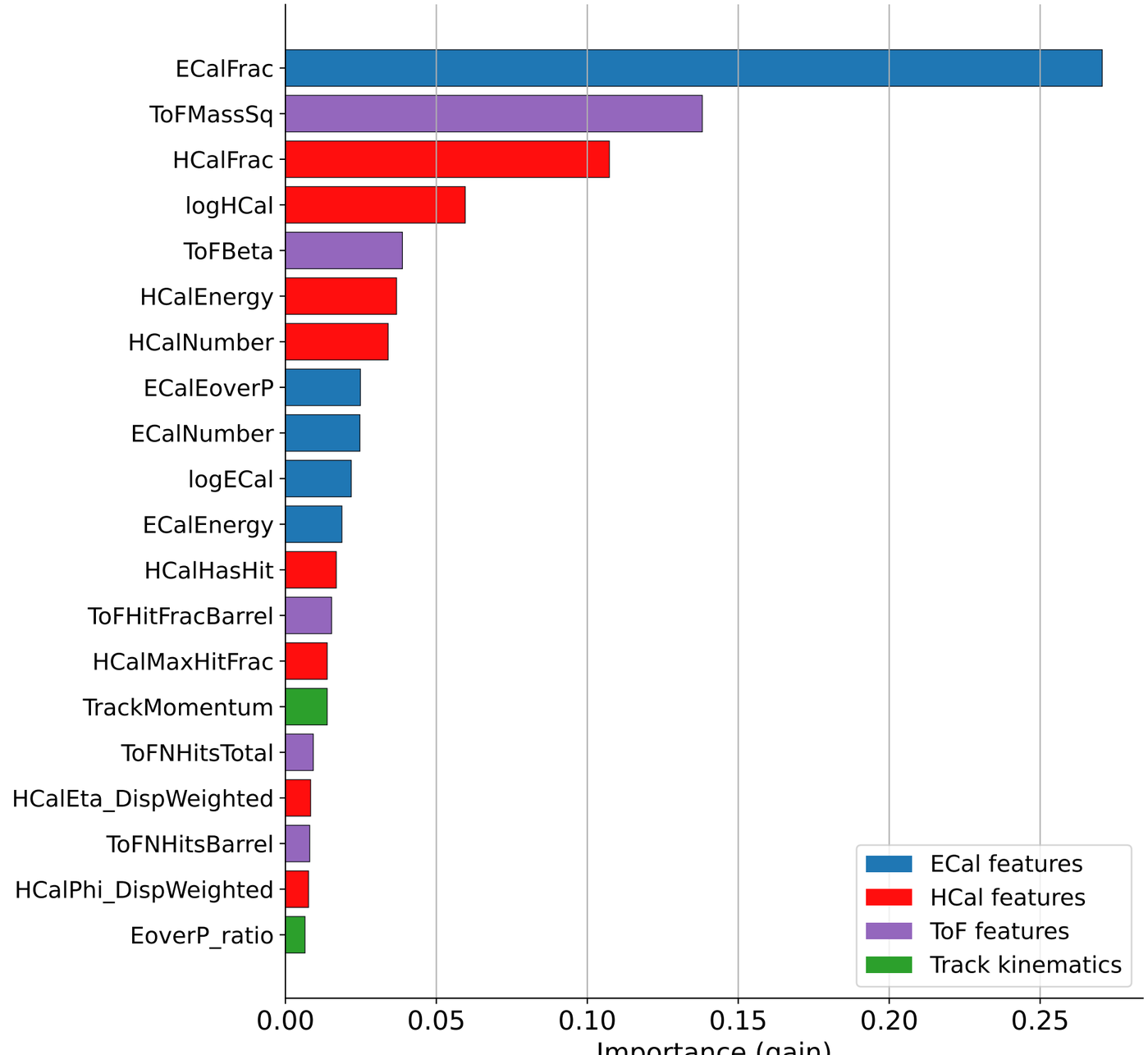
XGBoost Training Diagnostics



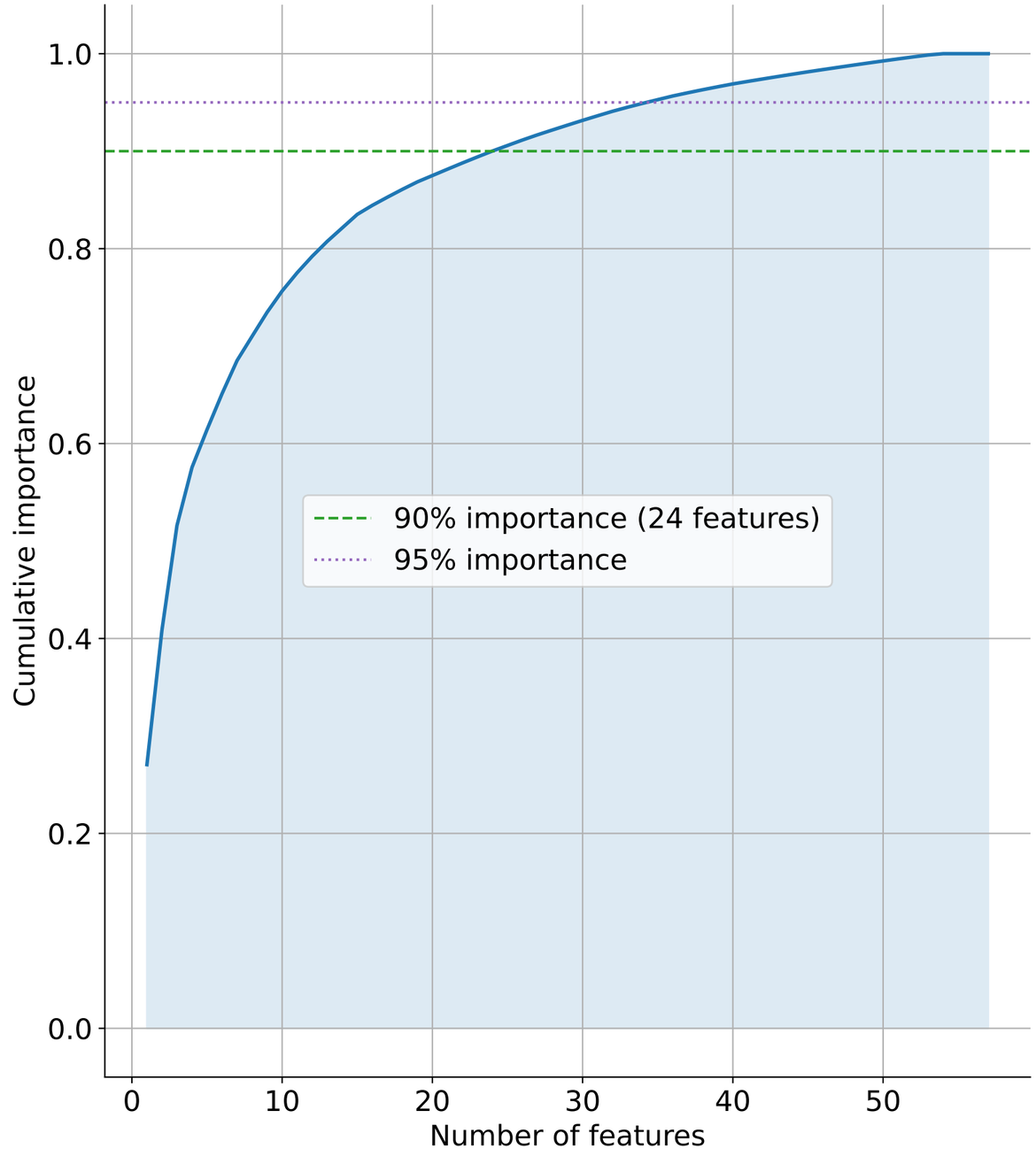


Feature Importances — XGBoost

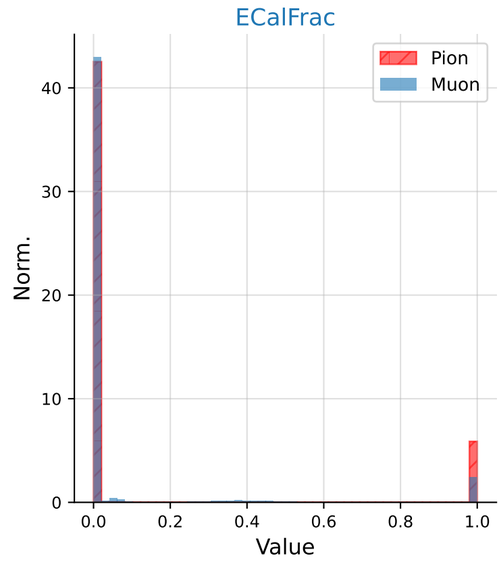
Top 20 Features

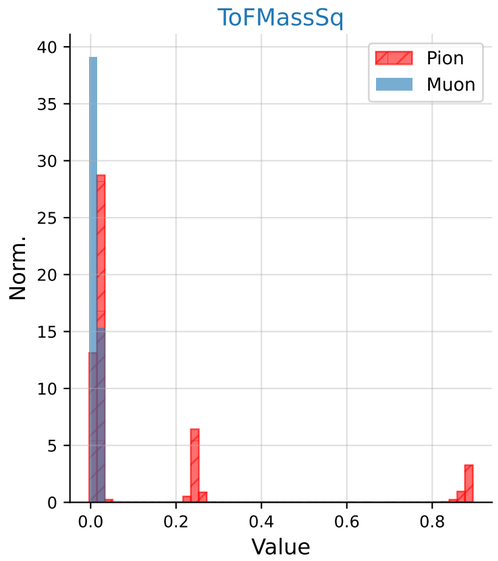


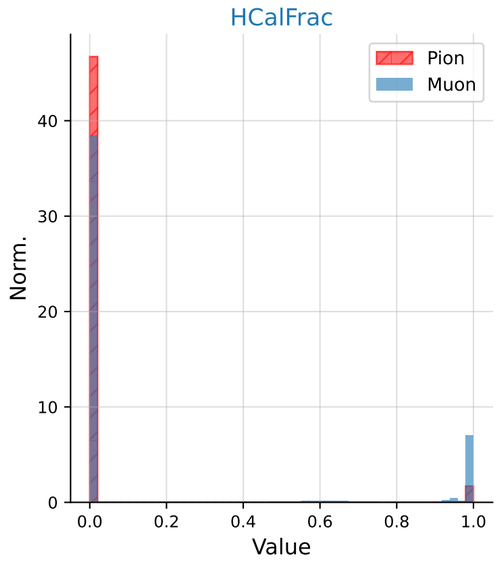
Cumulative Feature Importance

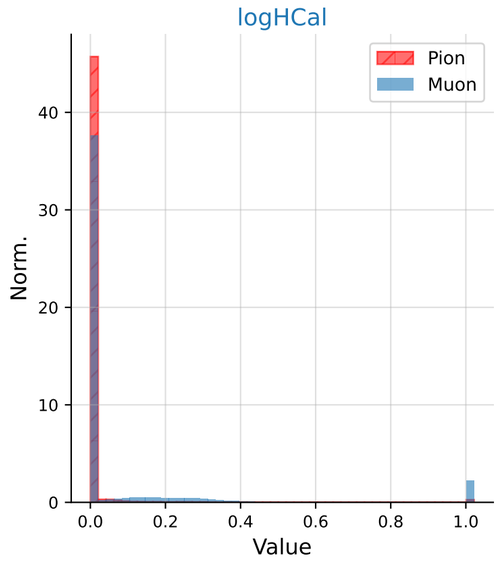


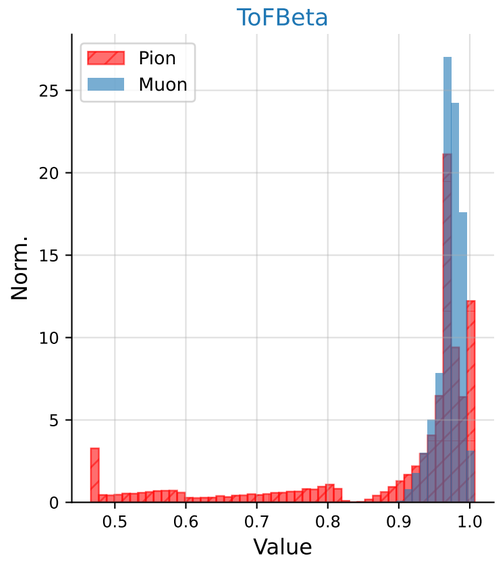
Key Physics Variable Distributions: Muon vs Pion
[full dataset before splitting]

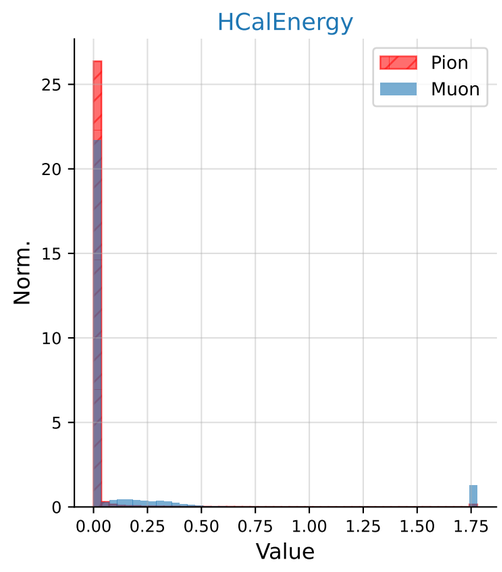


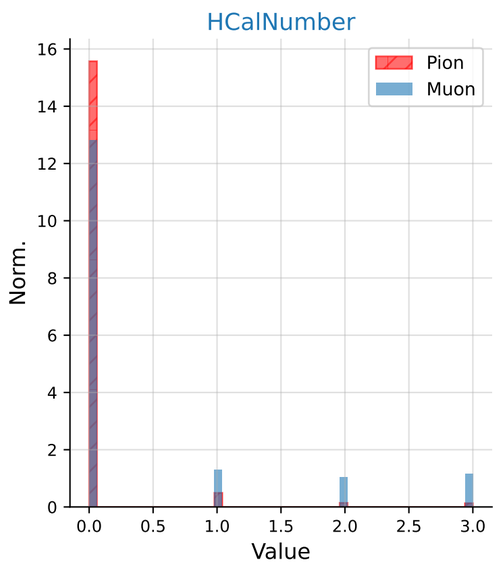


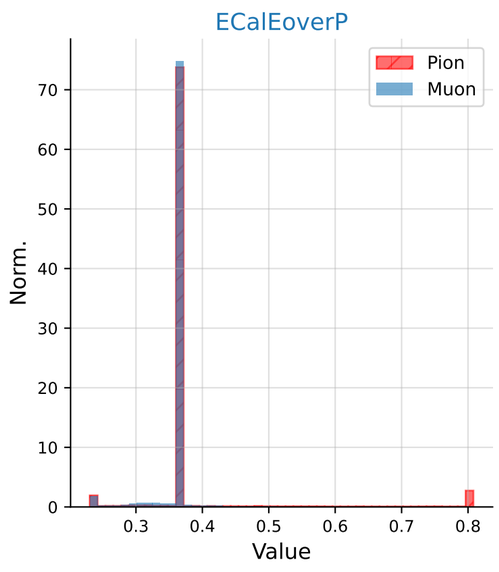


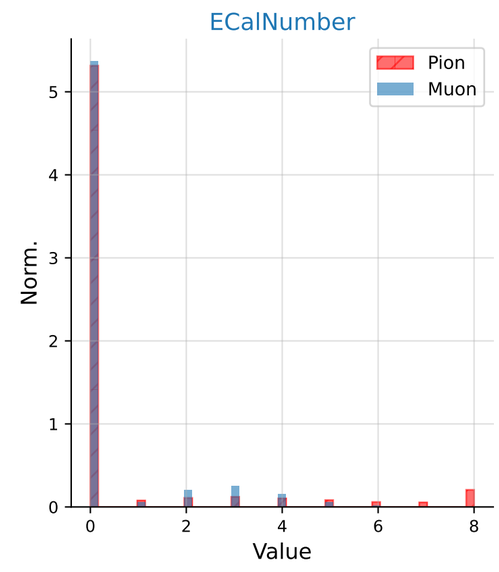


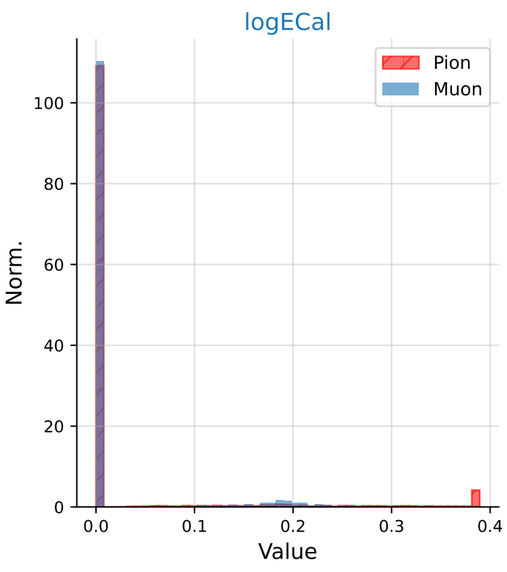


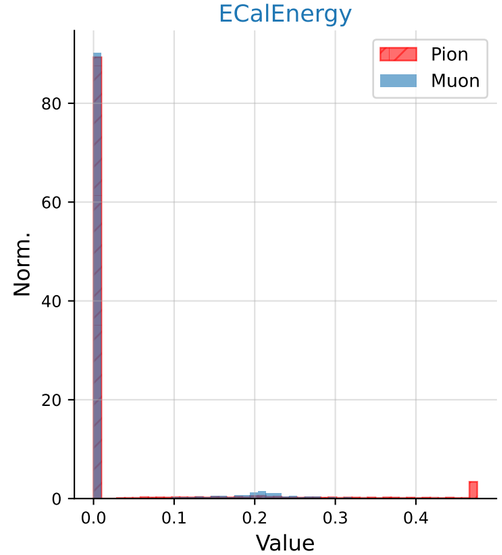


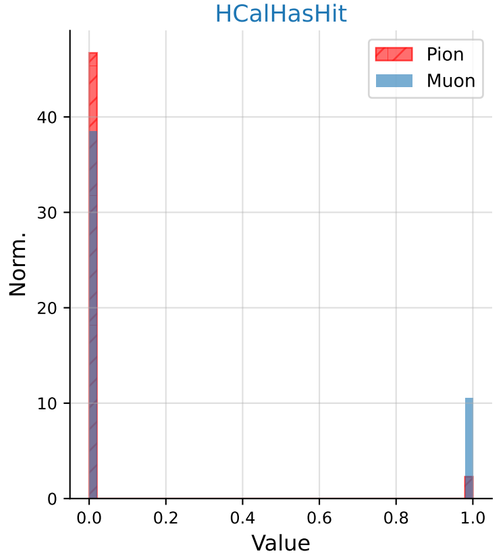


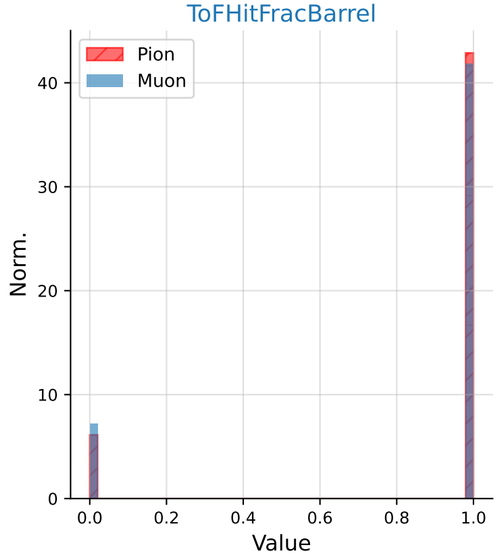


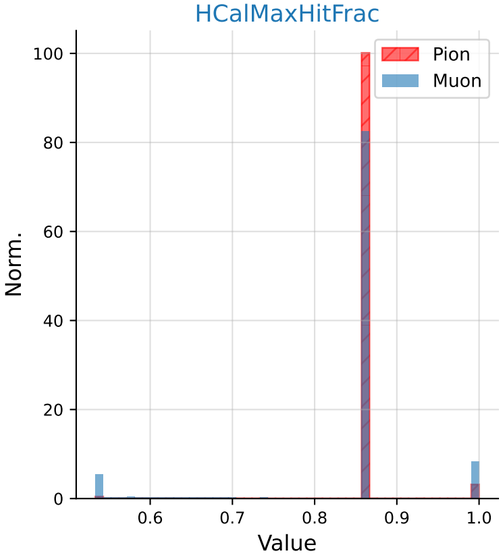


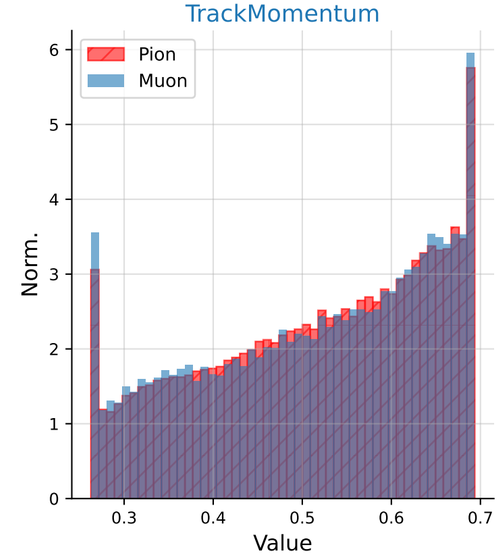












Spearman Correlation Matrix — All Features
(values from the full dataset)

Higher |ρ| means a stronger monotonic relationship between features

ECalEnergy
ECalNumber
ECalEoverP
ECalAvgHitEnergy
ECalSpreadPhi
ECalSpreadEta
ECalSpreadR
ECalMaxHitFrac
ECalEnergyStdDev
ECalEnergyConcentration
ECalR_Disp
ECalR_DispWeighted
ECalEta_DispWeighted
ECalPhi_DispWeighted
HCalEnergy
HCalNumber
HCalEoverP
HCalAvgHitEnergy
HCalSpreadPhi
HCalSpreadEta
HCalSpreadR
HCalMaxHitFrac
HCalEnergyStdDev
HCalEnergyConcentration
HCalR_Disp
HCalR_DispWeighted
HCalEta_DispWeighted
HCalPhi_DispWeighted
ECalHasHit
HCalHasHit
ECalFrac
HCalFrac
HitRatio
EoverP_ratio
logECal
logHCal
MaxHitFrac_mismatch
AvgHitEnergy_ratio
SpreadR_mismatch
EnergyConcentration_mismatch
R_DispWeighted_ratio
ToFBeta
ToFMassSq
ToFNHitsBarrel
ToFNHitsEndcap
ToFNHitsTotal
ToFMinDistBarrel
ToFMinDistEndcap
ToFAvgLenBarrel
ToFAvgLenEndcap
ToFHasToF
ToFHitFracBarrel
ToFMatchedSubdet
ToFMinDistCombined
TrackMomentum
TrackEta
TrackPhi

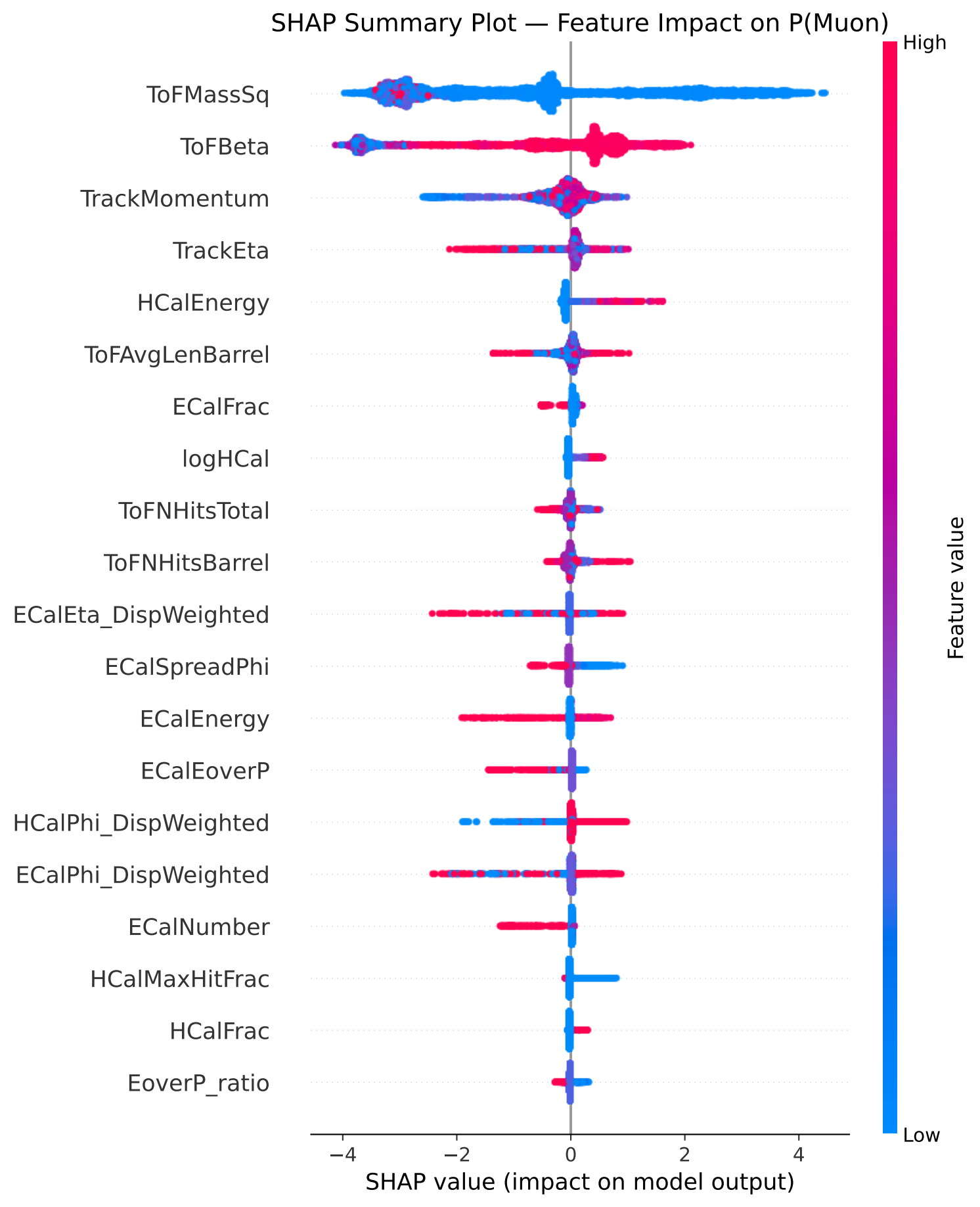
Spearman ρ
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

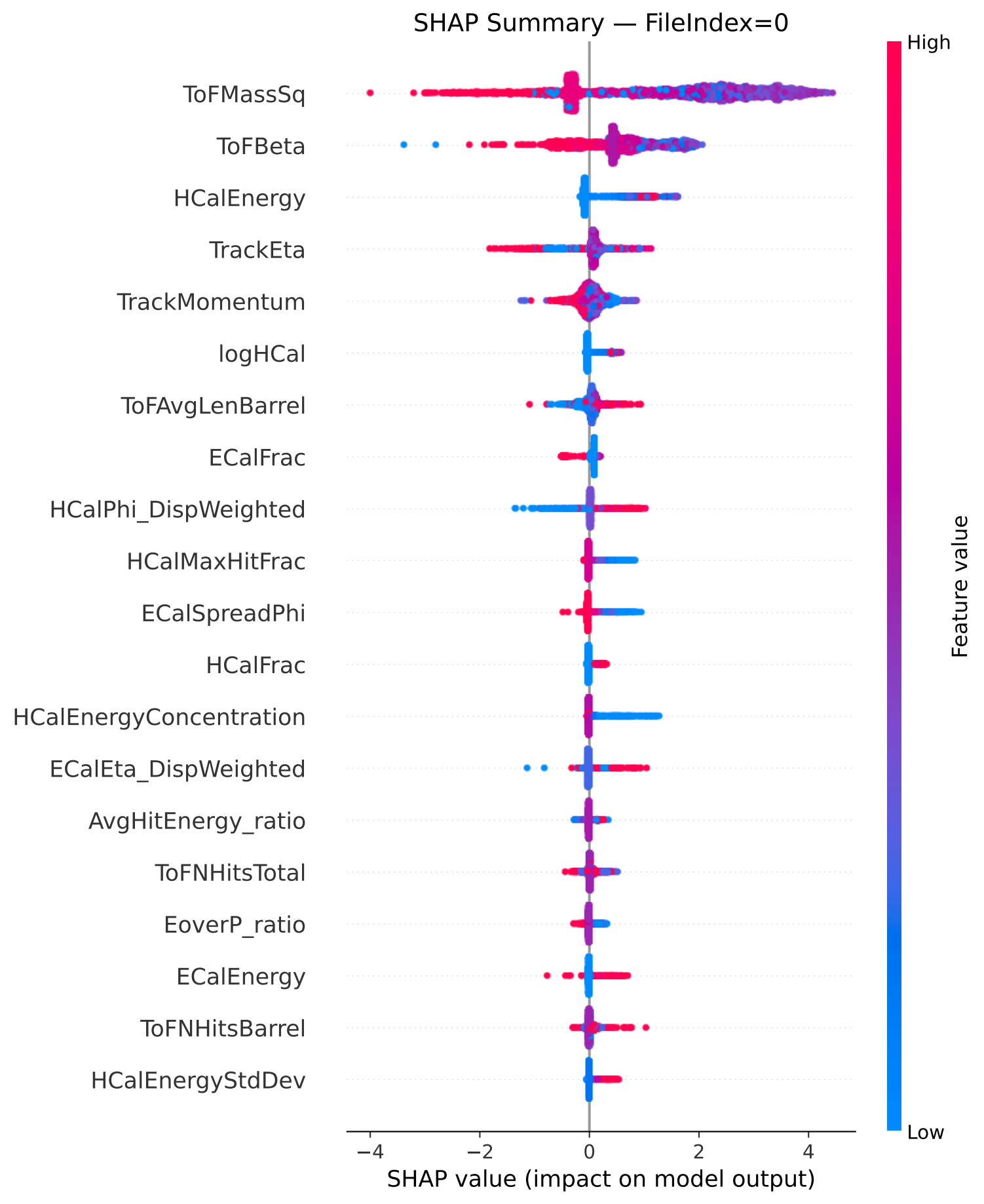
Feature correlation with muon/pion class
Blue = positive (→ muon), Red = negative (→ pion)

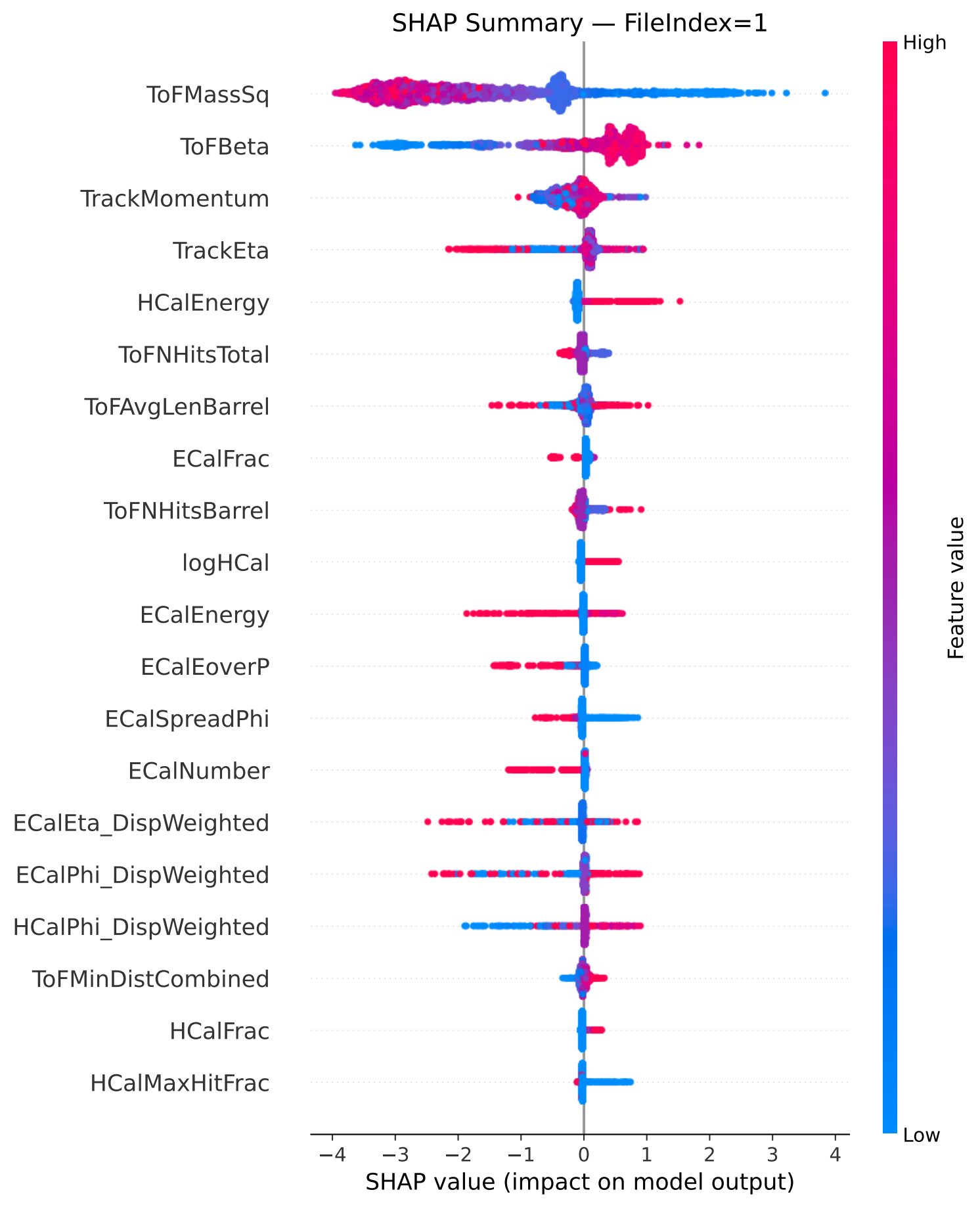
ToFMinDistEndcap
ToFNHitsEndcap
TrackMomentum
TrackPhi
TrackEta
ECalEta_DispWeighted
ECalR_DispWeighted
ToFAvgLenEndcap
AvgHitEnergy_ratio
ToFMinDistBarrel
ToFMinDistCombined
ECalHasHit
ToFAvgLenBarrel
ECalPhi_DispWeighted
HCalEta_DispWeighted
ECalMaxHitFrac
ECalEnergy
logECal
ECalNumber
ECalR_Disp
ECalFrac
HCalAvgHitEnergy
ToFMatchedSubdet
ToFHitFracBarrel
ToFHasToF
HCalPhi_DispWeighted
ToFNHitsBarrel
ToFNHitsTotal
ECalEnergyConcentration
ECalAvgHitEnergy
HCalEoverP
ECalSpreadR
ECalEnergyStdDev
HCalSpreadEta
SpreadR_mismatch
MaxHitFrac_mismatch
HCalEnergyStdDev
HCalSpreadPhi
EnergyConcentration_mismatch
HCalEnergyConcentration
HCalMaxHitFrac
ECalSpreadEta
EoverP_ratio
ECalEoverP
ECalSpreadPhi
HitRatio
R_DispWeighted_ratio
HCalR_Disp
HCalSpreadR
HCalR_DispWeighted
ToFBeta
HCalFrac
HCalHasHit
HCalNumber
logHCal
HCalEnergy
ToFMassSq

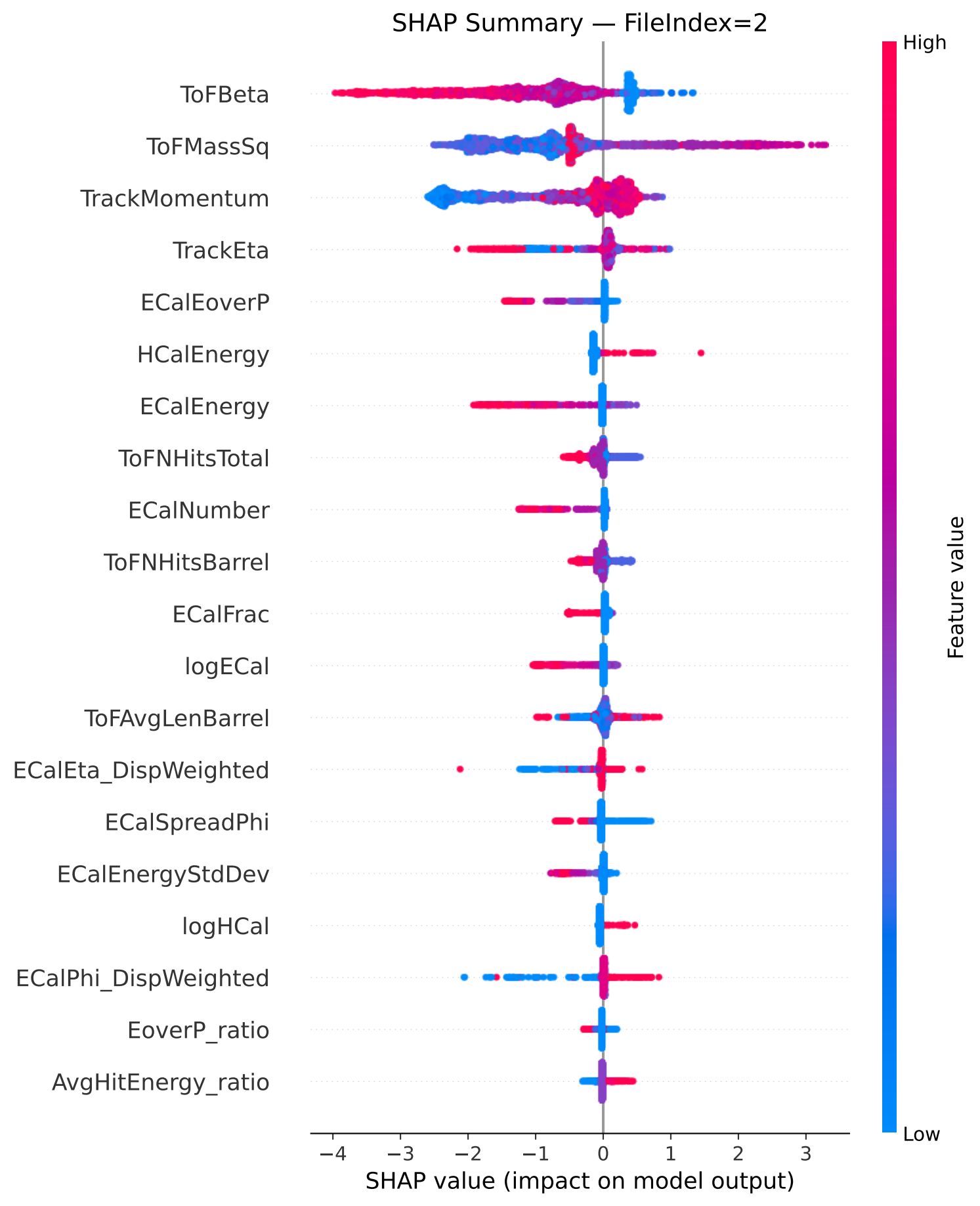
−0.4 −0.3 −0.2 −0.1 0.0 0.1 0.2

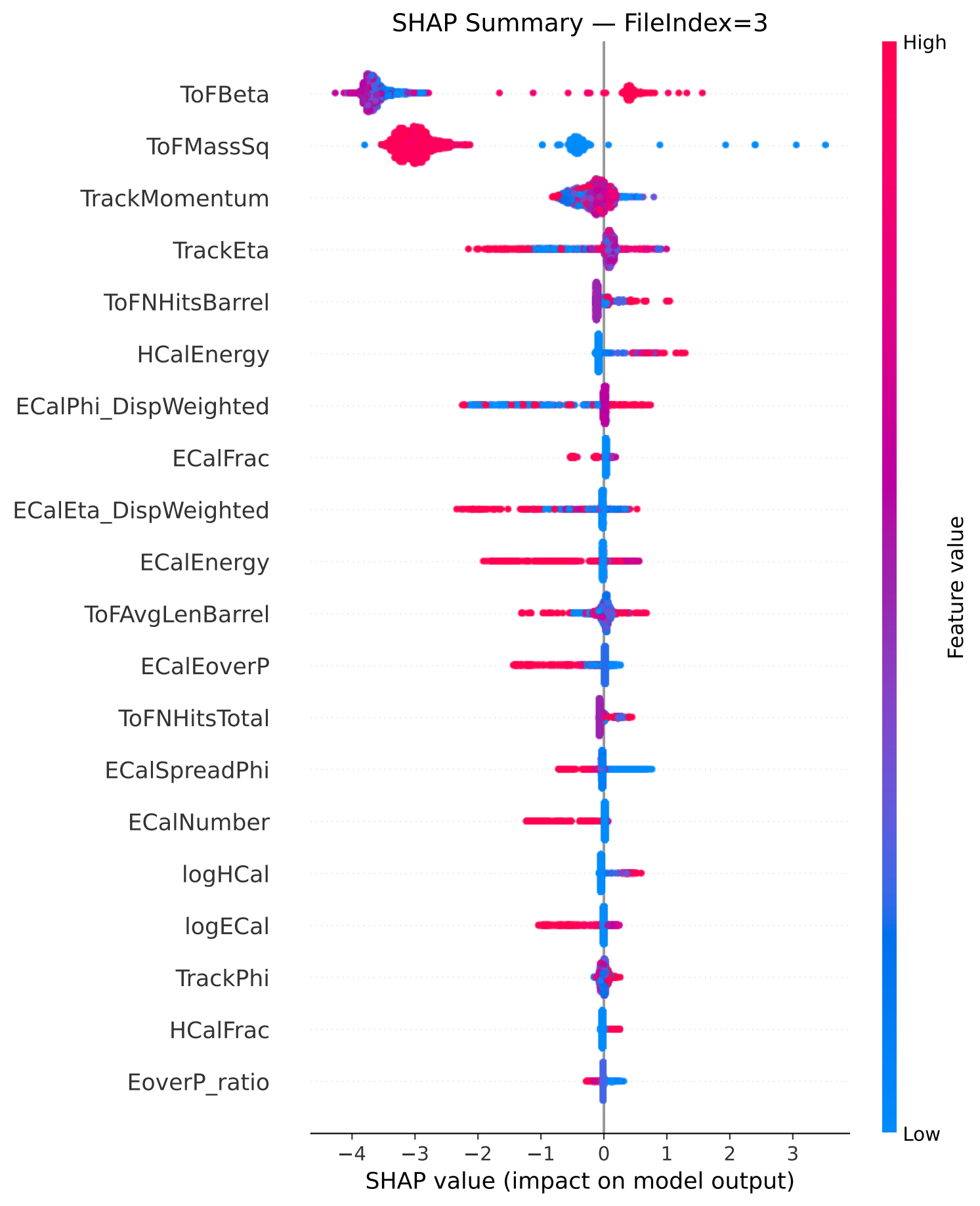
Spearman ρ with the label (IsMuon)

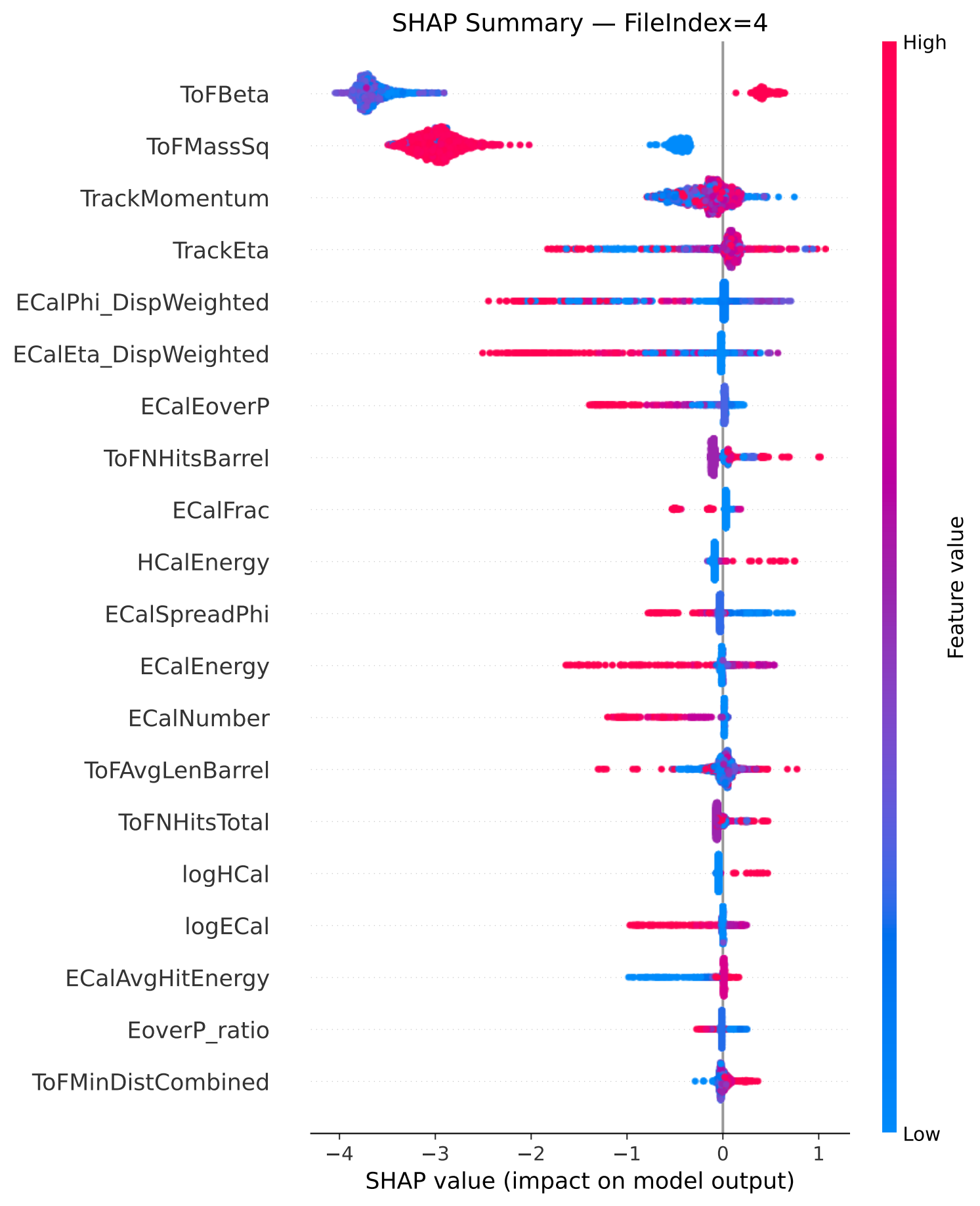

SHAP Summary Plot — Feature Impact on P(Muon)
ToFMassSq
ToFBeta
TrackMomentum
TrackEta
HCalEnergy
ToFAvgLenBarrel
ECalFrac
logHCal
ToFNHitsTotal
ToFNHitsBarrel
ECalEta_DispWeighted
ECalSpreadPhi
ECalEnergy
ECalEoverP
HCalPhi_DispWeighted
ECalPhi_DispWeighted
ECalNumber
HCalMaxHitFrac
HCalFrac
EoverP_ratio
−4
−2
0
2
4
SHAP value (impact on model output)
High
Feature value
Low

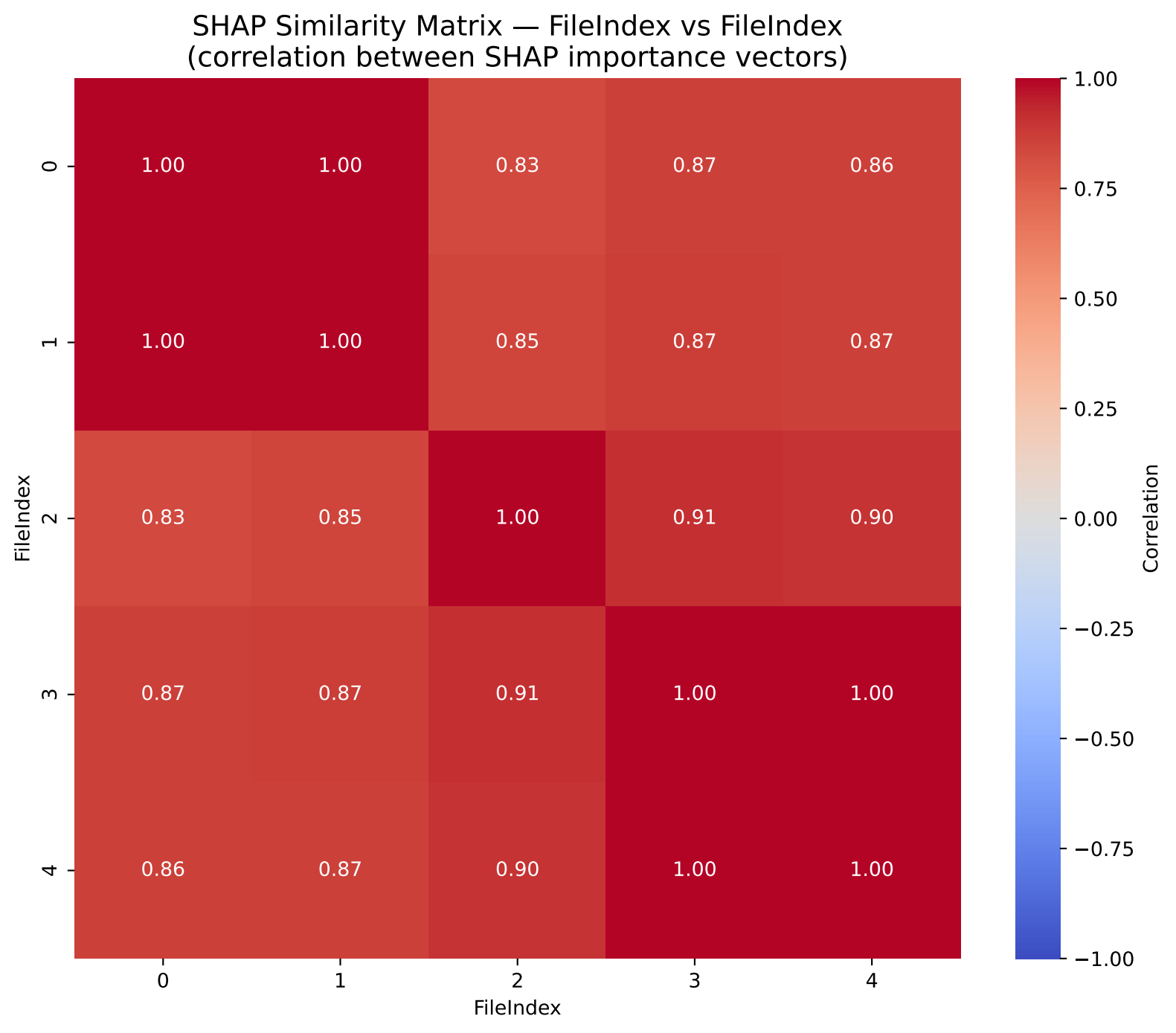

SHAP Summary — FileIndex=0
ToFMassSq
ToFBeta
HCalEnergy
TrackEta
TrackMomentum
logHCal
ToFAvgLenBarrel
ECalFrac
HCalPhi_DispWeighted
HCalMaxHitFrac
ECalSpreadPhi
HCalFrac
HCalEnergyConcentration
ECalEta_DispWeighted
AvgHitEnergy_ratio
ToFNHitsTotal
EoverP_ratio
ECalEnergy
ToFNHitsBarrel
HCalEnergyStdDev
−4
−2
0
2
4
SHAP value (impact on model output)
High
Feature value
Low


SHAP Summary — FileIndex=1
ToFMassSq
ToFBeta
TrackMomentum
TrackEta
HCalEnergy
ToFNHitsTotal
ToFAvgLenBarrel
ECalFrac
ToFNHitsBarrel
logHCal
ECalEnergy
ECalEoverP
ECalSpreadPhi
ECalNumber
ECalEta_DispWeighted
ECalPhi_DispWeighted
HCalPhi_DispWeighted
ToFMinDistCombined
HCalFrac
HCalMaxHitFrac
−4
−3
−2
−1
0
1
2
3
4
SHAP value (impact on model output)
High
Feature value
Low

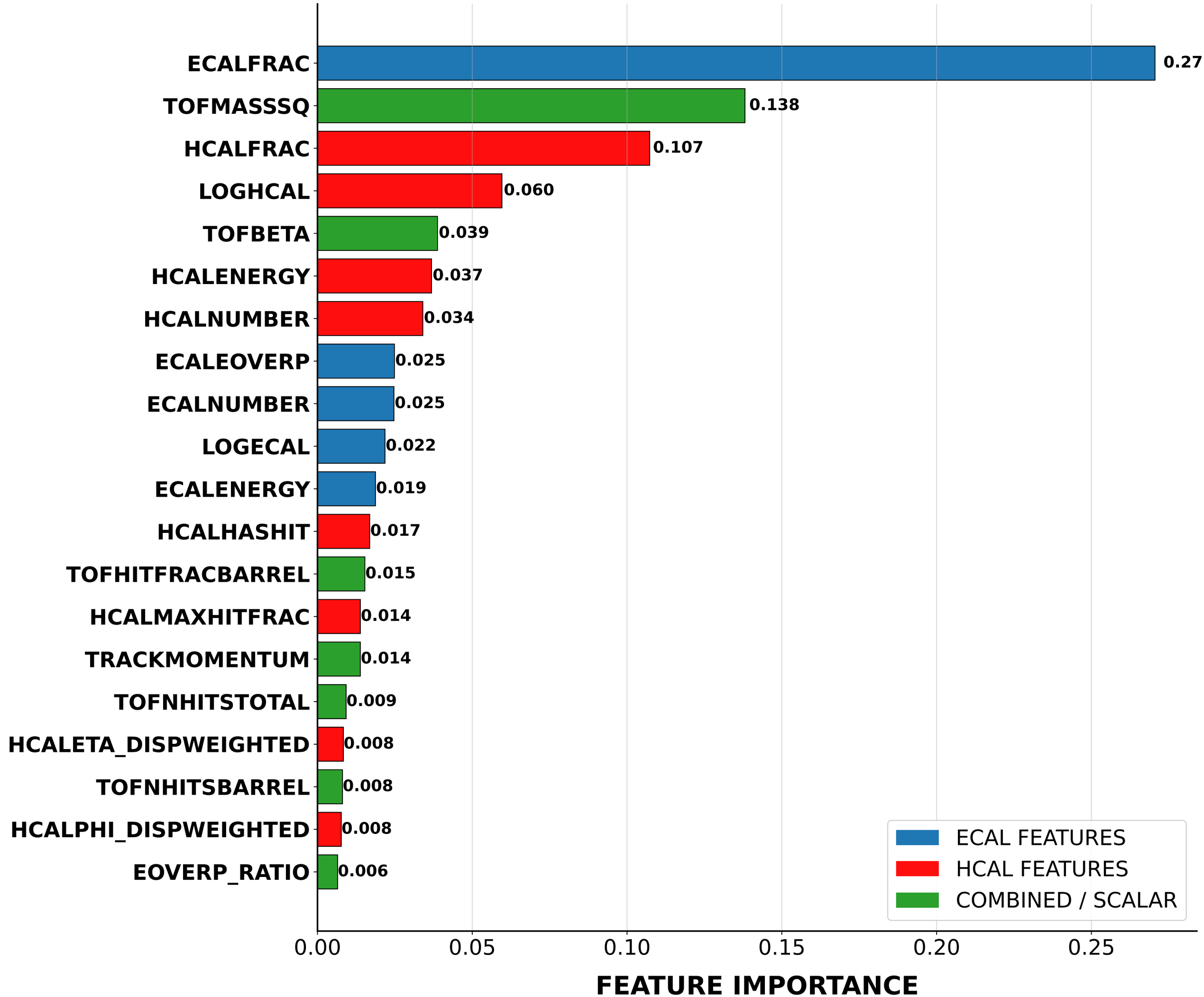

SHAP Summary — FileIndex=2
ToFBeta
ToFMassSq
TrackMomentum
TrackEta
ECalEoverP
HCalEnergy
ECalEnergy
ToFNHitsTotal
ECalNumber
ToFNHitsBarrel
ECalFrac
logECal
ToFAvgLenBarrel
ECalEta_DispWeighted
ECalSpreadPhi
ECalEnergyStdDev
logHCal
ECalPhi_DispWeighted
EoverP_ratio
AvgHitEnergy_ratio
−4
−3
−2
−1
0
1
2
3
SHAP value (impact on model output)
High
Feature value
Low


SHAP Summary — FileIndex=3
ToFBeta
ToFMassSq
TrackMomentum
TrackEta
ToFNHitsBarrel
HCalEnergy
ECalPhi_DispWeighted
ECalFrac
ECalEta_DispWeighted
ECalEnergy
ToFAvgLenBarrel
ECalEoverP
ToFNHitsTotal
ECalSpreadPhi
ECalNumber
logHCal
logECal
TrackPhi
HCalFrac
EoverP_ratio
−4
−3
−2
−1
0
1
2
3
SHAP value (impact on model output)
High
Feature value
Low

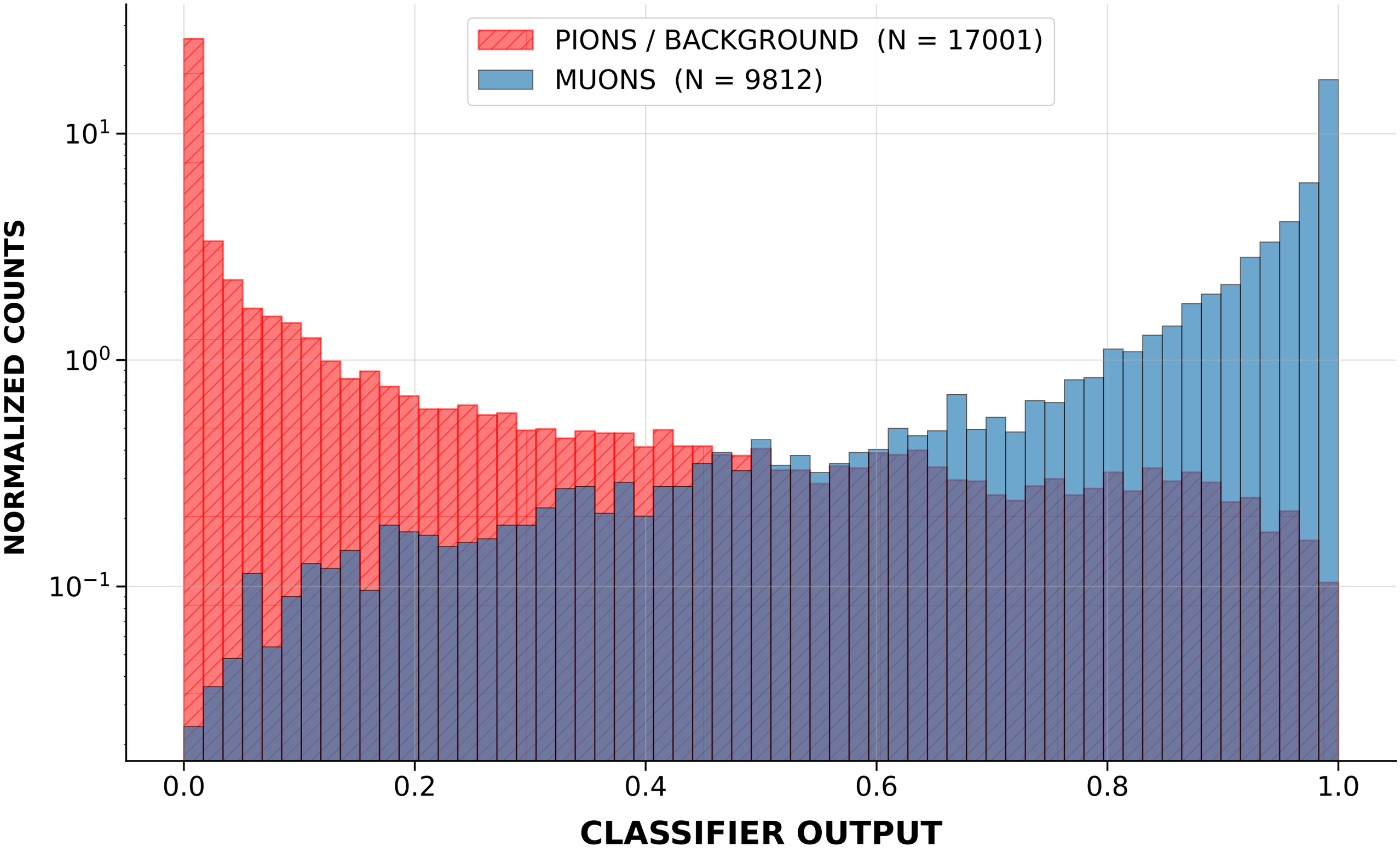

SHAP Summary — FileIndex=4
ToFBeta
ToFMassSq
TrackMomentum
TrackEta
ECalPhi_DispWeighted
ECalEta_DispWeighted
ECalEoverP
ToFNHitsBarrel
ECalFrac
HCalEnergy
ECalSpreadPhi
ECalEnergy
ECalNumber
ToFAvgLenBarrel
ToFNHitsTotal
logHCal
logECal
ECalAvgHitEnergy
EoverP_ratio
ToFMinDistCombined
−4
−3
−2
−1
0
1
SHAP value (impact on model output)
High
Feature value
Low


SHAP Similarity Matrix — FileIndex vs FileIndex
(correlation between SHAP importance vectors)
FileIndex
0
1
2
3
4
1.00 1.00 0.83 0.87 0.86
1.00 1.00 0.85 0.87 0.87
0.83 0.85 1.00 0.91 0.90
0.87 0.87 0.91 1.00 1.00
0.86 0.87 0.90 1.00 1.00
FileIndex
Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

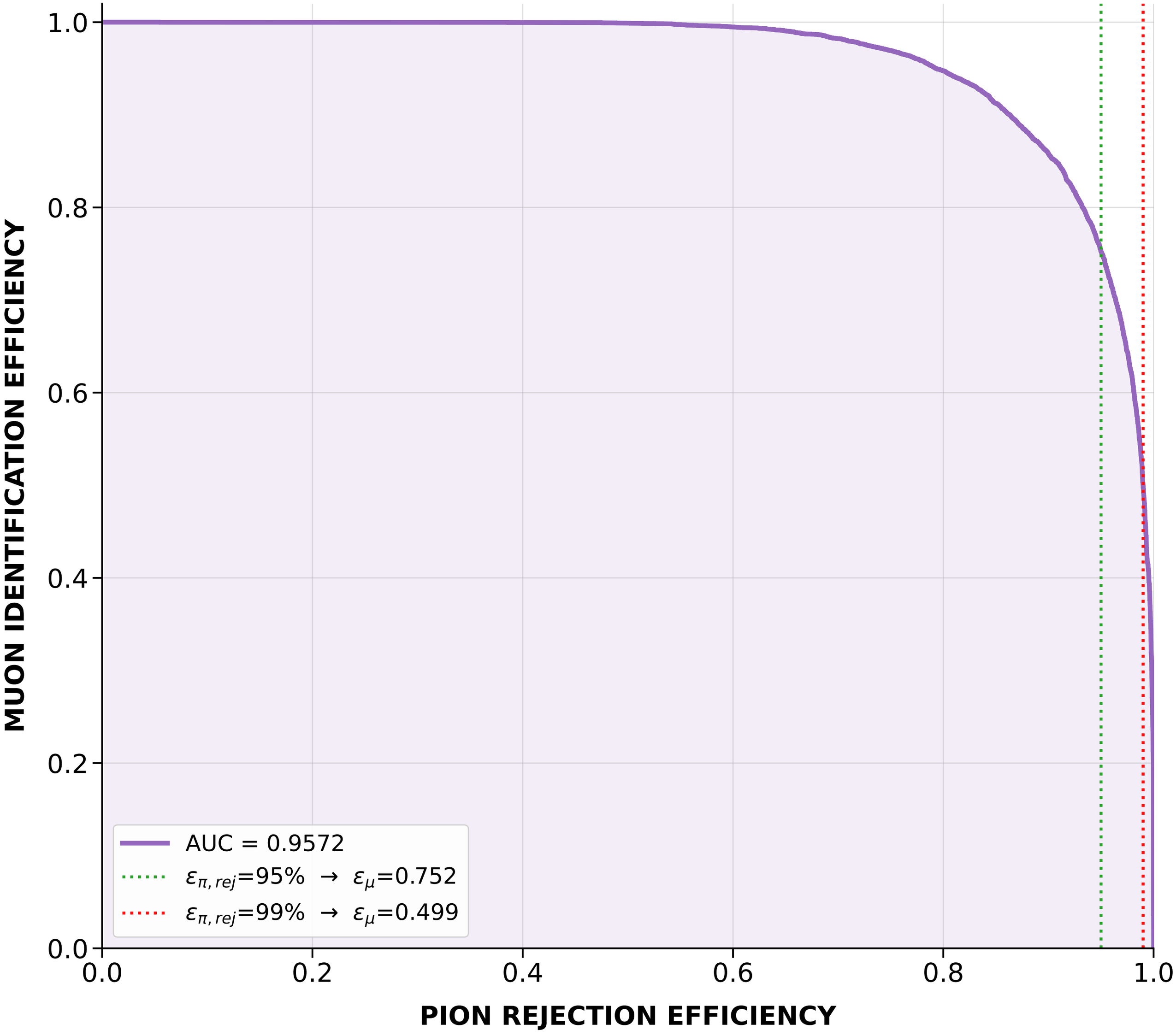
FEATURE IMPORTANCE — TOP 20



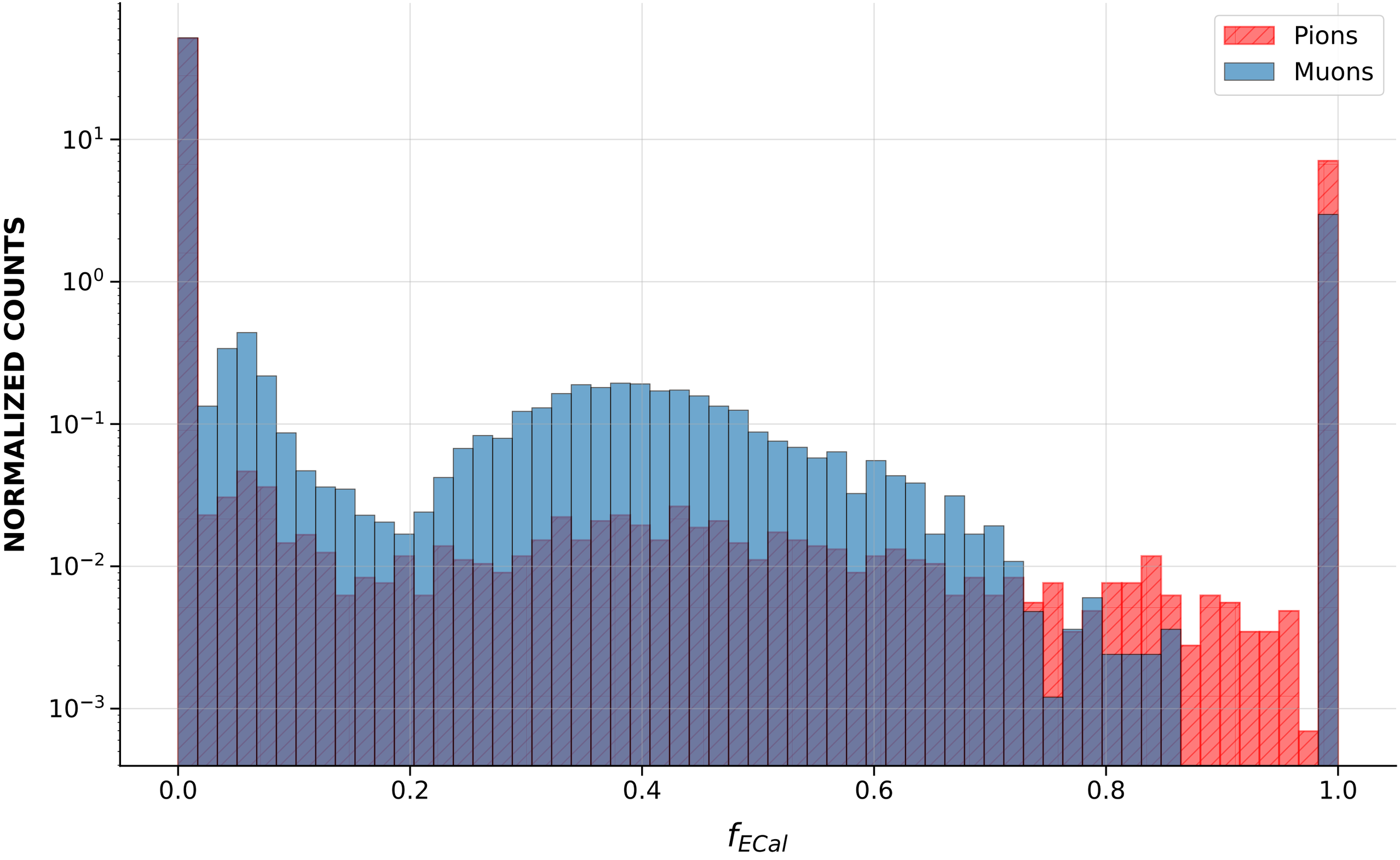
CLASSIFIER SCORE DISTRIBUTION
MUONS vs PIONS



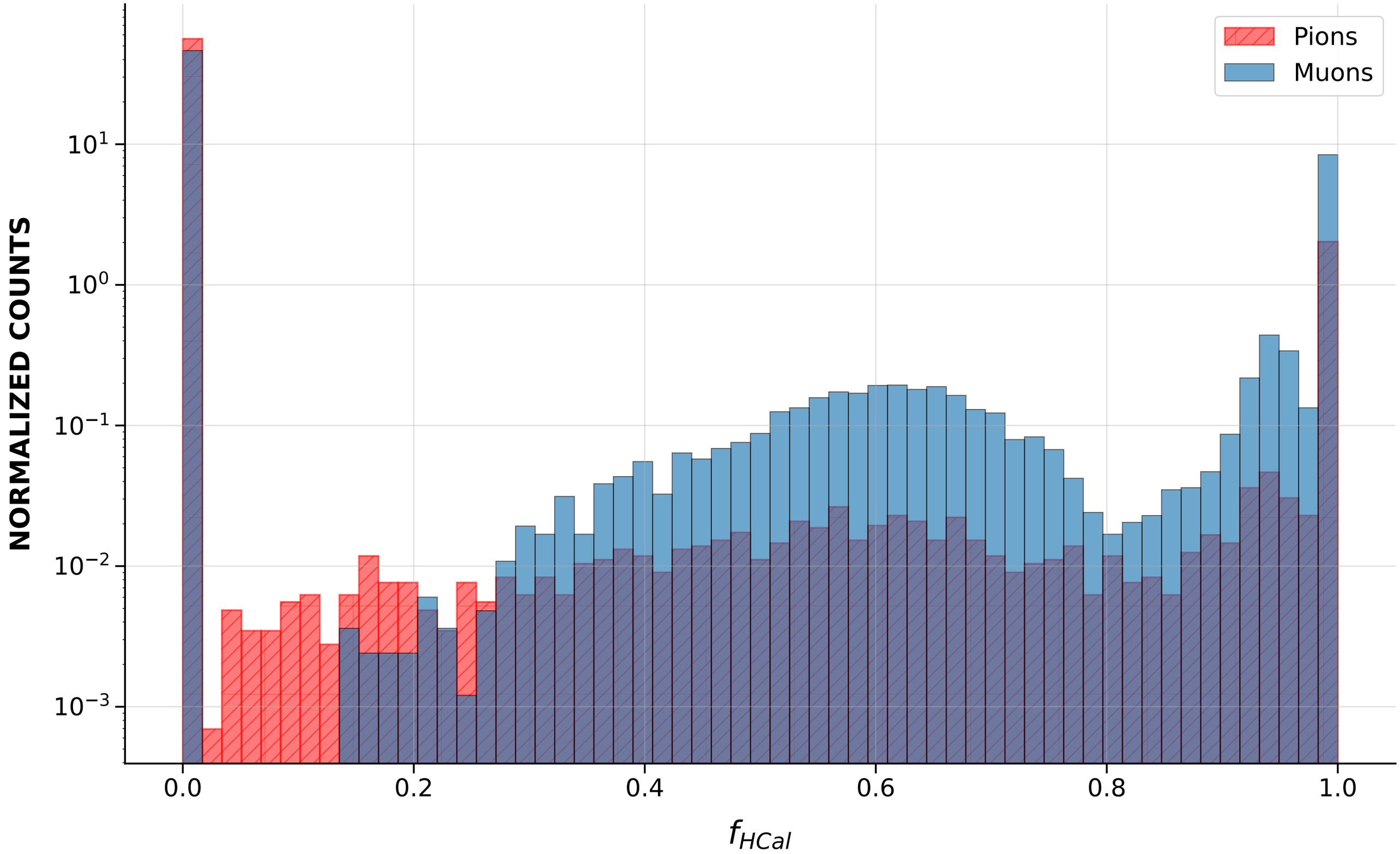
ROC CURVE: ε_μ vs $\varepsilon_{\pi,\text{rej}}$



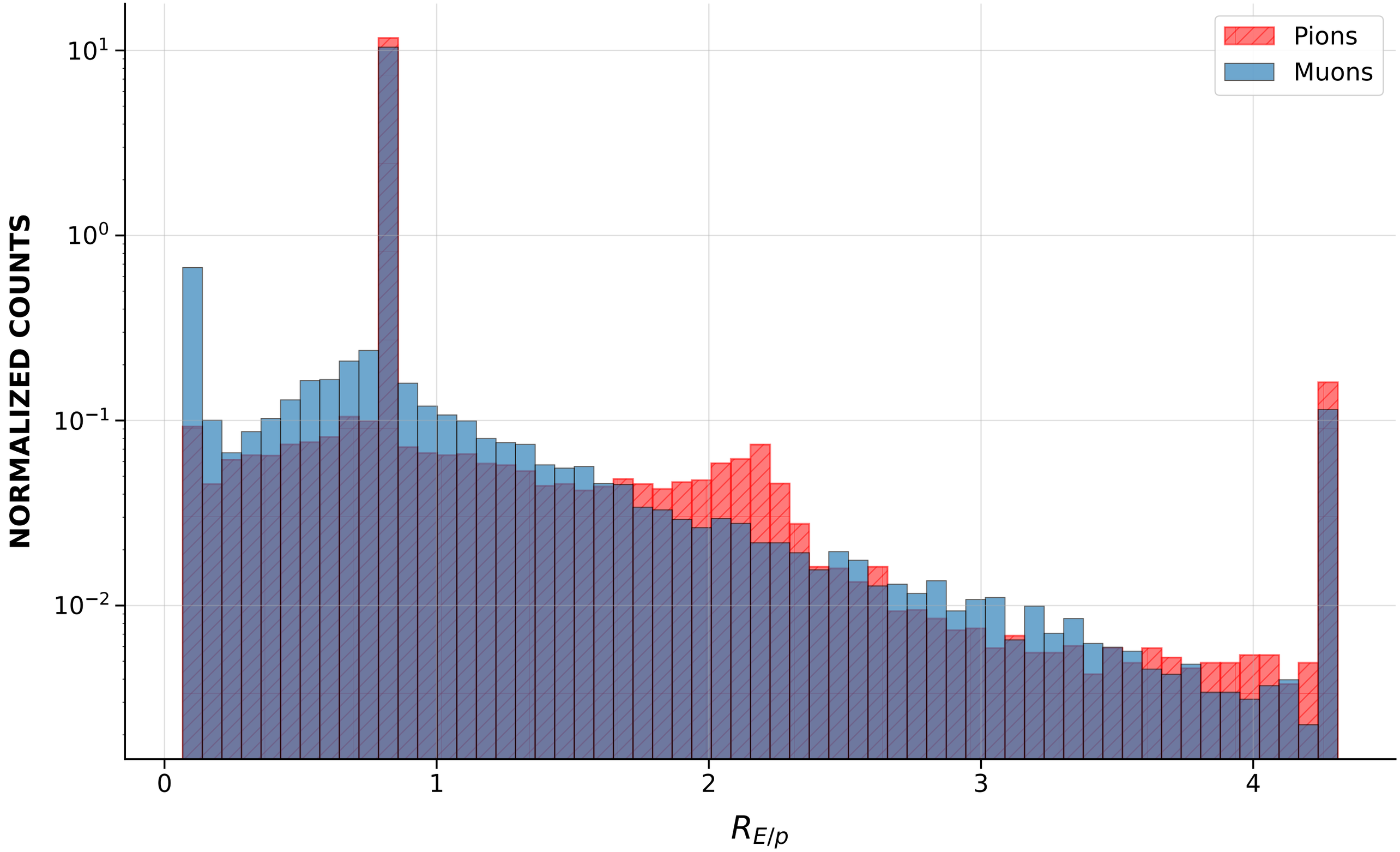
ENGINEERED CALORIMETER FEATURE
ECalFrac



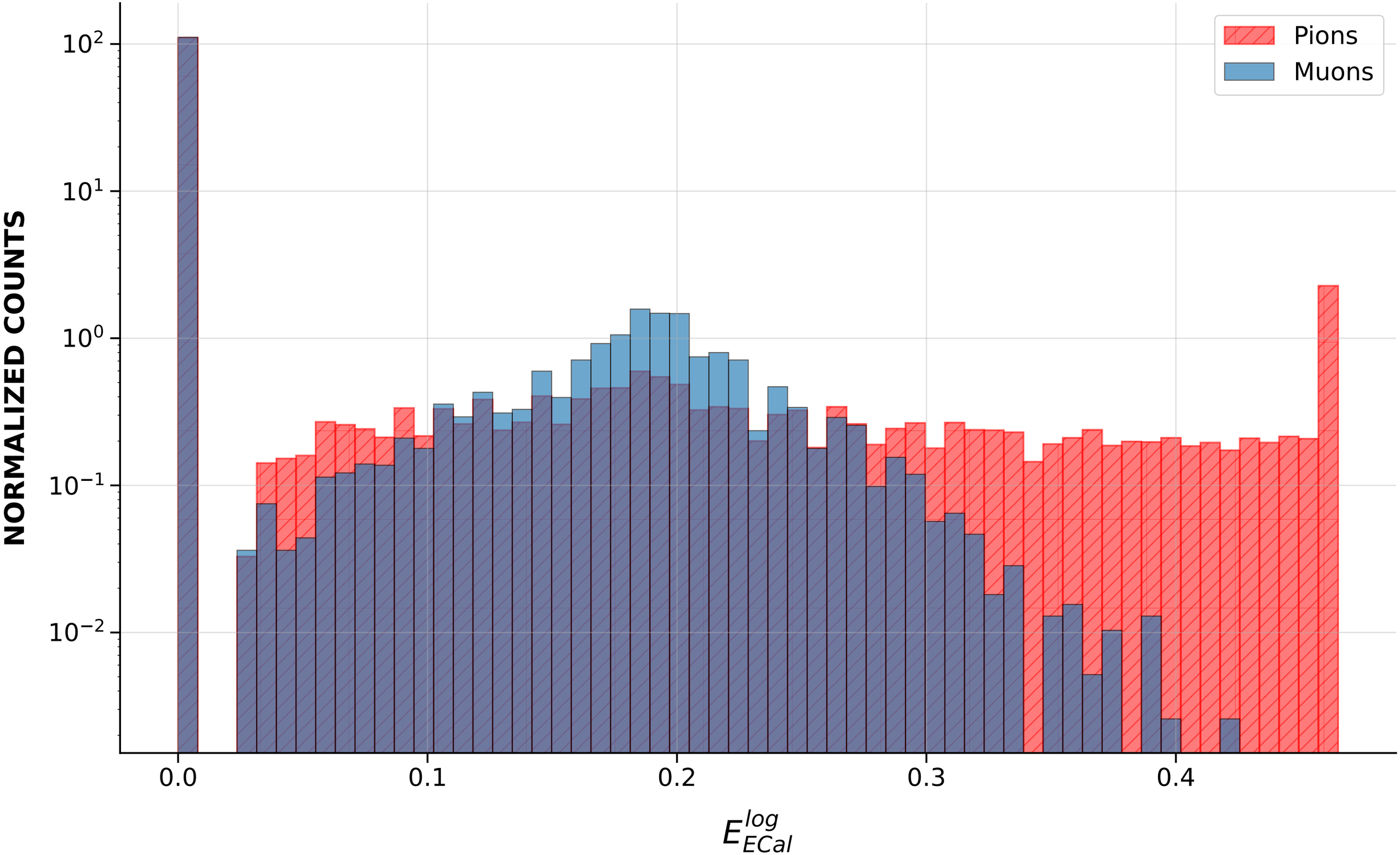
ENGINEERED CALORIMETER FEATURE
HCalFrac



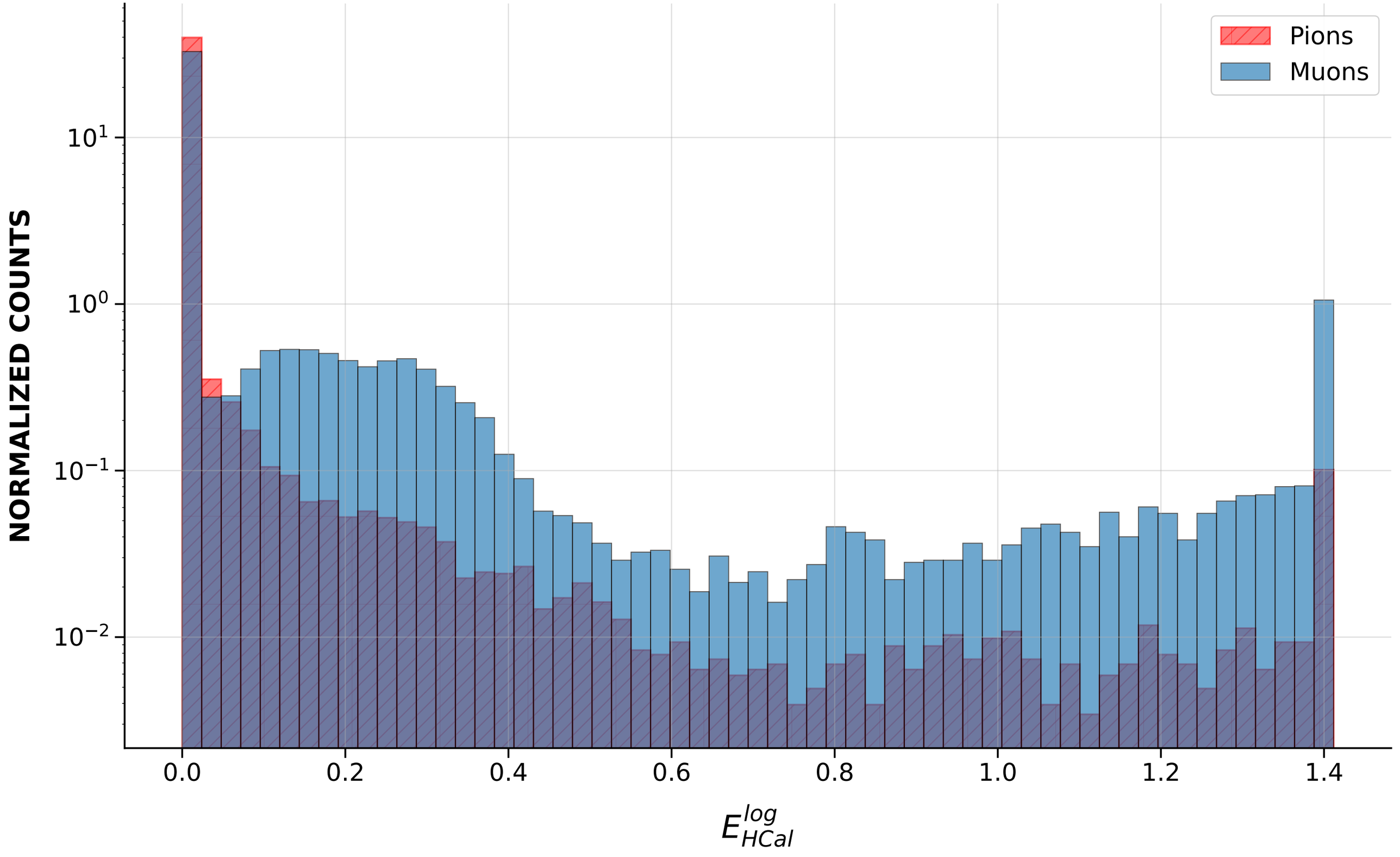
ENGINEERED CALORIMETER FEATURE
EoverP_ratio



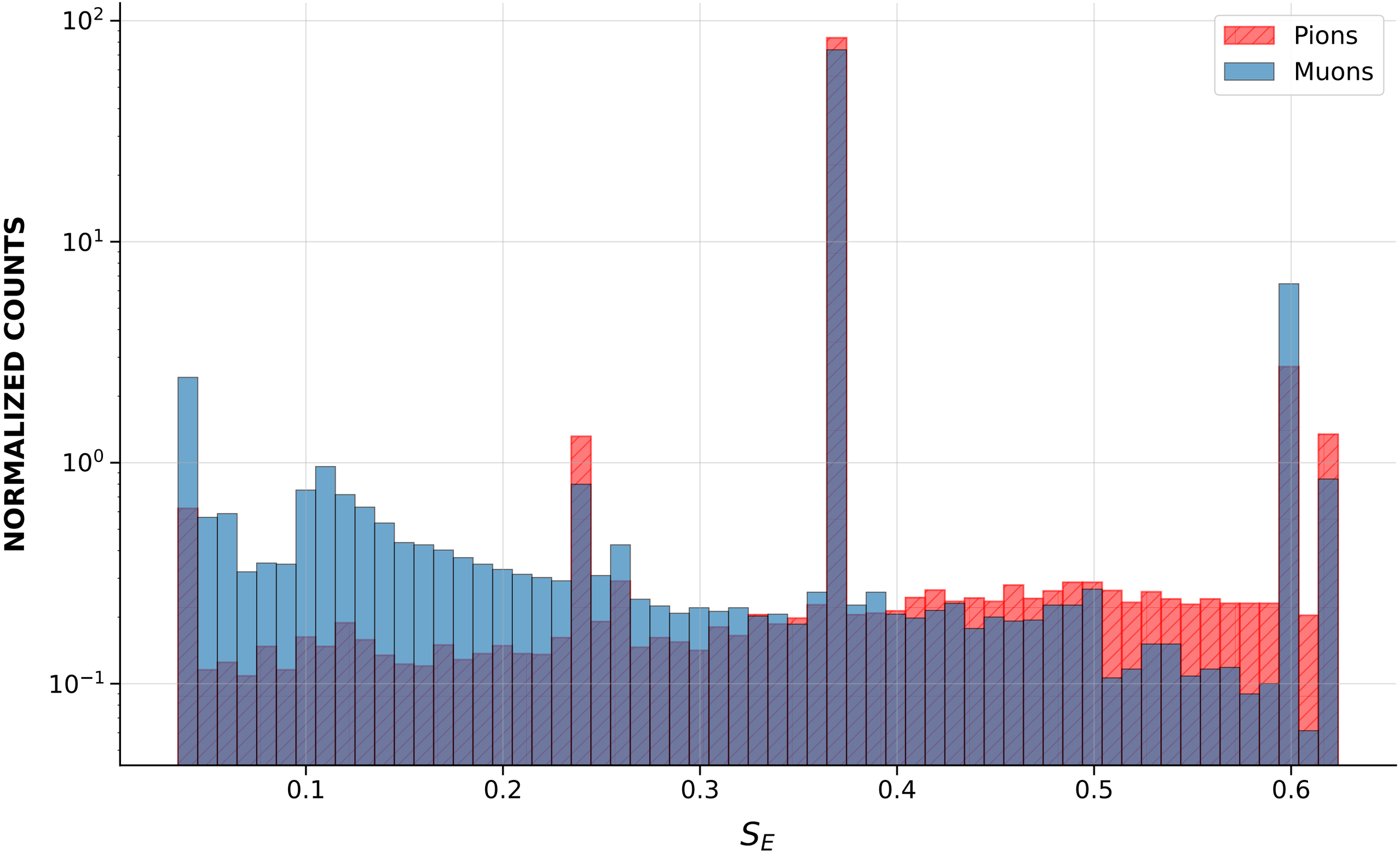
ENGINEERED CALORIMETER FEATURE
logECal



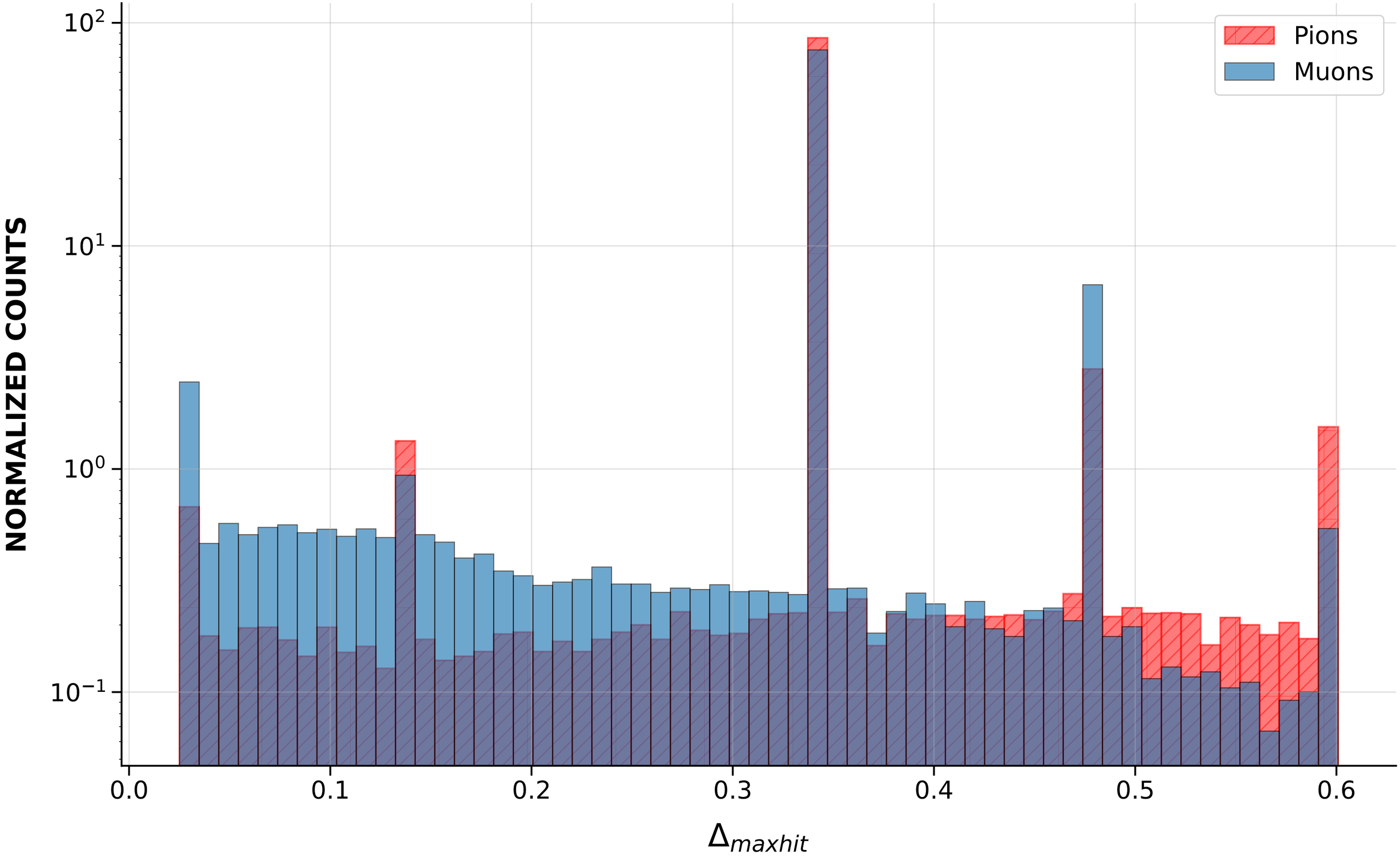
ENGINEERED CALORIMETER FEATURE
logHCal



ENGINEERED CALORIMETER FEATURE
EnergyConcentration_mismatch



ENGINEERED CALORIMETER FEATURE
MaxHitFrac_mismatch



ENGINEERED CALORIMETER FEATURE
HitRatio

